

## Abstract

**The general claim.** Process-based models are not self-justifying. A surplus-production model or a one-pool water balance is more mechanistic than last-value persistence, and that added mechanism is widely treated as a reason to prefer it. It is not: it is a *claim*, testable against a naive benchmark, and it can fail. The value of a process model is bounded by the predictability of the driver it persists. Where a model’s causal content consists of carrying a driver forward in time, and that driver is close to white, the structure adds parameters without adding signal — the mechanism is real but carries no forecast information at the horizon in question, and persisting it is overhead rather than insight. *A model that persists a near-white driver has no claim to beat persistence, whatever its mechanistic fidelity.* Two consequences are general: a negative result of this kind is informative only against a protocol fixed before any score is computed; and the failure is *diagnostic*, locating the defect in the driver’s spectrum rather than only reporting a loss.

**Why two systems.** A finding on one system is a demonstration; the same finding on two systems from different domains is a replication. This paper scores deliberately simple process-based forecast models against naive benchmarks on *two* systems under one frozen retention rule: the Northern cod spawning-stock biomass series (NAFO 2J3KL, 1983–2015), a marine fishery, and the J-17 annual-mean head series of the Edwards Aquifer (1934–2023), a karst groundwater system. The two have no hydrology, no biology and no institutional history in common. The mechanism by which added structure fails differs between them — a weakly autocorrelated surplus in the fishery, a near-white annual recharge in the aquifer — and that difference is the point: *the test is the same even though the mechanism is not.* Each system is held to the standard that added structure clears a naive benchmark, and the verdict is allowed to be negative.

**What is found.** On both systems, at the annual origin, last-value persistence is more accurate than the process-based alternatives scored. On the cod series no module is retained on either unpooled specification; on the aquifer the univariate AR(1) is retained, by a margin the paper shows to be a coin-flip. At multi-year horizons climatological baselines win on the aquifer, robustly. Under known ground truth the rule recovers a true autonomous module in 97% of replicates at collapse-window parameters, so the empirical non-retention of that module is evidence about the series rather than about the instrument — while for the stock-flow and depensation alternatives at recovery-window parameters power falls below 15%, so their non-retention is weak evidence. The two systems are never pooled: scores are reported separately throughout, and no cross-system average is taken.

**Structure.** Part I is the Northern cod study and Part II the Edwards Aquifer study. Each is self-contained — its own data, specification, models, results, discussion and conclusions — and each is reproduced in full from its companion paper without condensation. The method they share (the locked retention rule and the evaluation design) is restated in each part with the system-specific detail it carries there. A cross-system conclusion follows.

# Forecasting under a locked retention rule: process models, naive benchmarks, and two systems

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# Part I — Northern cod (*Gadus morhua*), NAFO 2J3KL

**Source study:** Does a surplus-production ladder improve forecasts of Northern cod? A scored test on NAFO 2J3KL. September 24, 2026.

Prepared in the format of Fisheries Research

## Highlights

- A five-module structural ladder is scored against two naive baselines
- No module attains lower pooled rolling-origin RMSE than persistence
- From a pre-collapse origin, every model misses the 1991–1995 collapse
- Simulation: the rule is specific, and powerful only where signal is strong
- Non-retention is strong evidence for the autonomous module, weak for others
- Outcome years 2016–2021, threshold definition-dependence (29 of 71 years), and the 2022–2023 survey suspension: the verdict layer, not forecast structure, moved

## Abstract

Five surplus-production modules and two naive baselines are scored on Northern cod (*Gadus morhua*), NAFO 2J3KL, under a rule whose scoring core was fixed beforehand: a module is kept only if it lowers RMSE against the next-simpler module and persistence. The predictand is NCAM (Northern Cod Assessment Model) M-shift SSB (DFO, 2016, Table A2), 1983–2015, LRP (limit reference point) 884.6 kt.

No module is retained on these two unpooled series, on rolling RMSE pooled over origins within each specification and horizon: under the coarse pass one-year RMSE is 98 kt for persistence against 115–196 kt, five-year 265 against 289–488 kt. By period the ordering is not uniform. From the fixed pre-collapse origin every model misses the 1991–1995 collapse (694–819 kt against 670 and 688 kt for baselines): constant productivity with a 1992 catch drop cannot reproduce it. Added structure is not uniformly harmful, yet none closes the gap. A second, unpooled specification (xteNCAM, 1954–2024, LRP 276 kt) gives the same outcome (matched persistence 84 kt against M1’s 120).

Simulation under known ground truth retains a true autonomous module in 97% of replicates at collapse-window parameters but has power below 15% for the stock-flow and depensation alternatives; against truth outside the tested model class the rule over-retains, so its specificity is conditional on that class. No Specification A margin against persistence has an interval excluding zero, though several on Specification B do. The predictand is retrospectively reconstructed rather than the vintages at each origin, and catch is supplied along the horizon: a conditional hindcast, not an operational forecast.

Published outcome data now cover 2016–2021: the outcome window is an increase regime (340–440 kt on the updated vintage, read within 1–26 kt by the independent xteNCAM vintage), persistence errs low throughout, and the window contains no collapse event, so the frozen verdict is neither confirmed nor reversed by it. On the published files the two reference points disagree on 29 of 71 years, the 1980s-mean LRP itself moves from 884.6 to 790.0 kt across vintages, and the real 2022–2023 survey suspension — two assessment cycles with the leading indicator uncomputable and removals frozen — supplies the review-timing boundary case. The decision-relevant uncertainty of the period accumulated in the observation and reference-point layer, not in forecast structure.

**Keywords:** northern cod; biomass forecasting; surplus production; forecast evaluation; stock assessment; prediction skill; model identifiability

## 1. Introduction

Fisheries assessment builds structure by default. State-space age-structured assessments, surplus-production models, and ecosystem-linked extensions each add parameters, state

variables, or couplings in the hope of better advice. In the surplus-production implementations evaluated here, that structure does not pay for itself: across a five-module ladder scored on two unpooled assessment specifications, no tested implementation attains lower pooled rolling-origin RMSE than last-value persistence at either evaluated horizon. Split by period the pooled result is not uniform — persistence is not the lowest-error rule during the collapse band — and Section 3.5 reports that split. The result is scoped to the estimators and series tested, and the exercise is a conditional hindcast rather than an operational forecast test. Whether that structure improves out-of-sample forecasts — as opposed to in-sample fit — is a separate, testable question, and on it the burden of proof sits with the added structure.

That burden is now formalised in assessment practice. Comparison against a naive baseline has become a standard acceptance diagnostic: hindcast cross-validation scores a model’s out-of-sample predictions with the mean absolute scaled error (MASE) of Hyndman and Koehler (2006), which divides prediction error by the error of a random-walk forecast, so that a value above one means the model predicts less accurately than assuming no change (Kell et al., 2016; Kell et al., 2021). The diagnostic cookbook of Carvalho et al. (2021) lists prediction skill so measured as one of four criteria for accepting an integrated assessment, alongside convergence, fit, and retrospective consistency, and the good-practice guidance for surplus-production models adopts the same test (Kokkalis et al., 2024). The reference forecast in that literature is therefore already the one used here.

Two features of that practice leave the present question open. First, MASE is almost always computed on the *abundance index* a model fits, not on the estimated stock state that advice is written about, so a model can pass on the index it was tuned to while its biomass trajectory remains untested. Second, the diagnostic is applied to certify a single accepted assessment rather than to rank a graded sequence of structural additions against each other and against the naive rule at once. What is missing is a direct, scored comparison in which the predictand is the assessment’s own biomass output, each increment of structure has a declared comparator, and the naive rule is a competitor rather than a scaling constant. That is the comparison reported here.

Northern cod (*Gadus morhua*) in NAFO divisions 2J3KL is the canonical demanding case — and the running instance of this paper. The late-1980s and early-1990s collapse — a spawning stock falling from a high, weakly trending phase to a few percent of its 1980s mean — occurred under an assessment and management regime whose failures were dissected in the canonical post-mortems (Hutchings and Myers, 1994; Walters and Maguire, 1996). The non-recovery that followed the July 1992 moratorium has kept the depensation question open for three decades (Shelton and Healey, 1999). A partial rebuilding was documented in the 2010s (Rose and Rowe, 2015), but surplus production has since stalled, with some years being negative (Rose, 2026).

The measured quantity throughout is the condition of the productive stock — the spawning-stock biomass whose persistence underwrites future yield. This is distinct from the catch taken from it. A fishery can return a high extracted yield while the stock that produces it declines, and it is the latter that the forecast here must track. Any model family proposed for this stock must therefore first clear a minimal bar: it must at least beat the forecast that nothing changes. One-dimensional autonomous surplus-production maps carry a structural constraint under sustained constant catch, discussed with its fitted values and its scope in Section 4. It is a property of a map held at a fixed removal level, not a reason the forecasts below fail, and the scored comparison does not isolate it.

This evaluation follows a stated retention rule whose scoring core was coded before the first scoring pass and applied unchanged (Section 4 records its protocol status): additional model structure is retained only when it reduces primary RMSE relative to the next-simpler model and relative to persistence (§2.3). Early-warning and intervention-selection criteria are secondary objectives; neither is operationalised here (the Brier score is reported as a secondary diagnostic and never changes a retention verdict, and the moratorium is deliberately not evaluated).

The instrument is a forward-ordered ladder of surplus-production models (not a strict nesting for M2 and M4) evaluated against two naive baselines. It runs output-only, stock-and-flow, residual, then lagged initialisation; prey- and index-informed observation variants are evaluated separately in Section 3.4 and are not part of the ladder. A module is kept only if it improves the stated score.

The paper asks whether stock-flow, residual, delay, or prey-informed modules reduce SSB forecast error relative to last-value persistence and to the autonomous surplus-production model. The scope is deliberately narrow: this paper does not estimate an Allee threshold, identify the cause of the 1990s collapse, or evaluate whether the 1992 moratorium was adequate.

The stock was selected because of its data availability (a full state-space assessment series), its policy relevance, and the sharpness of its collapse-and-stall trajectory. The same scored design is applied to a groundwater system, the Edwards Aquifer, Texas, in Abaee (2026, Does a one-pool water-balance model improve forecasts of Edwards Aquifer head). The two systems' series are not pooled, and no retention outcome is transferred between them.

The rule that decides retention is itself evaluated. Because a negative result is only as informative as the instrument that produced it, Section 3.7 reports a pre-registered simulation of the rule's operating characteristics under known ground truth: how often it retains a module that is genuinely present, how often it retains structure when persistence is the truth, and how it compares with selection by an information criterion or by scaled error alone. That evaluation is what licenses the reading of the empirical result module by module rather than as a single verdict.

The remainder of the article is organized as follows. Section 2 states the data, the two assessment specifications, the model ladder, and the evaluation design. Section 3 reports the results: the primary specification (3.1), annual landings and the survey-start variant (3.2), the alternative assessment specification (3.3), prey-informed productivity (3.4), the Diebold–Mariano and block-bootstrap the Diebold–Mariano and block-bootstrap uncertainty analysis (3.5), the fitted parameters (3.6), and the simulated operating characteristics of the retention rule (3.7). Section 4 discusses the findings, the identification limits, the reconstruction-level corroboration, and the protocol status. Section 5 concludes. The contributions are: the scored forecast-ladder comparison on the full record — the surplus-production ladder against naive and assessment-based competitors under the declared evaluation design (Section 3.1), with the landings, survey-start, alternative assessment, and prey-informed variants (Sections 3.2–3.4); the uncertainty analysis of the score differentials — the Diebold–Mariano tests and the block-bootstrap, with the fitted parameters reported (Sections 3.5–3.6); and the retention rule's simulated operating characteristics with the delayed-revelation reading of the documented one-year lead (Section 3.7).

## Related work

**Testing forecast accuracy.** Comparing the predictive accuracy of competing forecasters on the same realisations, rather than comparing in-sample fits, is the standard discipline for this question (Diebold and Mariano, 1995), and the scaled error measures used below follow the established treatment of accuracy metrics (Hyndman and Koehler, 2006). The design here departs from the common practice of that literature in one respect: the retention rule — keep a module only if it lowers RMSE against both the next-simpler module and persistence — was fixed before any score was computed, and the rule itself is then evaluated by simulation under known ground truth.

**Guarding against selection after the fact.** Why a protocol must be fixed beforehand, and why stating it as a commitment rather than a preference matters, is the data-snooping problem: when many candidate forecasters are scored on the same realisations, the best observed score

is upward-biased as an estimate of skill, and the usual per-comparison tests do not restore validity. White’s reality check constructs the distribution of the best performance statistic across candidates under the null that none of them has skill (White, 2000), and the stepwise testing literature formalises the correction with control of the familywise error rate (Romano and Wolf, 2005).

The design here addresses the same hazard by a different and more restrictive route, and the difference should be stated rather than left implicit. Those procedures take a *set of already-computed scores* and ask which, if any, survive correction for having been selected. Here nothing is selected from the scores at all: the retention rule is a predicate fixed before any score is read, applied without exception, and its operating characteristics are measured by simulation under known ground truth. That is why a negative result can be reported as a finding rather than as an absence of evidence — the rule either retains a module or it does not, and on both systems it does not. The cost of the choice is scope, and it is a real cost: a rule fixed in advance cannot be tuned to the series, so a module that a more permissive protocol would have retained is still reported here as not retained.

**Simple benchmarks are hard to beat, which is the finding this paper shares.** That naive and simple statistical benchmarks are not far from the most accurate methods, and that sophisticated methods frequently fail to beat them, is the most consistently replicated result in forecasting. It is the organising finding of the M-competitions: in the largest of them, every one of the six submitted pure machine-learning methods performed poorly, none being more accurate than the combination benchmark and only one being more accurate than the naive benchmark, while the most accurate entries were combinations and hybrids of statistical methods (Makridakis, Spiliotis and Assimakopoulos, 2018, 2020).

The result below — no surplus-production module is retained against last-value persistence on either specification — is that phenomenon in a fisheries setting, and it should be read as a single-system contribution to it rather than as a surprise. **The scope difference is material and is not claimed away:** the M-competitions aggregate over tens of thousands of series and 61 forecasting methods, whereas this paper scores five modules plus two baselines on one stock under two specifications. A cross-series result about the average performance of method classes does not entail anything about a particular stock, and a single-stock result does not generalise. What the two share is the conclusion that added structure must be *earned* against a naive benchmark, and here it is not.

**Why persistence is hard to beat on this stock.** The collapse-window result — that constant productivity with a 1992 catch drop cannot reproduce the 1991–1995 collapse from the pre-collapse origin, with every model overshooting — belongs to the extensive literature on this stock’s collapse and non-recovery (Myers, Hutchings and Barrowman, 1997; Hutchings, 2005). That literature establishes the empirical fact of collapse and stalled recovery; this paper’s contribution is different in kind, namely to measure what that fact does to *forecast skill* from a fixed origin. A model cannot forecast a regime change it is not parameterised to represent, and the power analysis reported below quantifies the consequent limit: the rule retains a true autonomous module in 97% of replicates at collapse-window parameters, but has power below 15% against the stock-flow and depensation alternatives.

**What is not claimed.** This is a conditional hindcast: the predictand is the retrospectively reconstructed series rather than the vintage available at each origin, and catch is supplied along the horizon. It is therefore not an operational forecast evaluation, and the retained-versus-rejected verdicts reported here do not license claims about the performance of the assessment process as it actually ran.

## 2. Material and methods

### 2.1 Data and specifications

The specification tables type each field as follows: D, data; E, empirical construct; M, model; N, normative threshold.

**Table 1.** Primary specification — Specification A (the 1983–2015 NCAM M-shift SSB series of DFO, 2016, Table A2; the state-space assessment model of Cadigan, 2016).

| Field               | Contents                                                                                                   | Type |
|---------------------|------------------------------------------------------------------------------------------------------------|------|
| System              | Northern cod, NAFO 2J3KL, as represented by NCAM M-shift SSB                                               | D    |
| Interest            | Continuity of a spawning stock on the 2016 precautionary-approach cautious or healthy side of the 2016 LRP | N    |
| Domain              | Stock area in DFO (2016) Figure 1; calendar years 1983–2015                                                | D    |
| Safe set            | $S_t \geq \text{LRP} = 884.6 \text{ kt}$ (1983–1989 mean of Table A2)                                      | N    |
| Disturbance         | Unspecified productivity shocks; not a fitted $M(t)$                                                       | M    |
| Theoretical catch   | Any $C_t \geq 0$                                                                                           | M    |
| Implementable catch | Pre-1992 directed fishery; post-2 July 1992 moratorium and low inshore removals                            | E    |
| Horizon             | Hindcast 1983–2015; two fixed test windows and rolling origin                                              | D    |
| Food web            | Capelin excluded from the primary pass; tested as a variant in Section 3.4                                 | M    |
| Norm                | 2010/2016 DFO precautionary-approach LRP (1980s mean SSB), not the 2023 40% $B_{\text{MSY}}$ LRP           | N    |

Table 1 note: the safe-set and LRP fields type the reference frame of the evaluation; the primary RMSE score never uses them (only the secondary Brier indicator score does).

Table A2 also reports fishing mortality  $F$  and natural mortality  $M$ . They are joint outputs with SSB and are not used as exogenous inputs.

Regular et al. (2025) extend the assessment to 1954, revise the LRP to 276 kt (95% interval 180–423 kt; 40% of  $B_{\text{MSY}}$ ), and estimate 2024 SSB at 342 kt. That is a second specification — Specification B (the 1954–2024 extended xteNCAM series; Section 3.3). The two objects differ in five specification fields: (1) the state equation (xteNCAM versus NCAM M-shift), (2) the start year (1954 versus 1983), (3) the reference threshold (276 kt versus 884.6 kt), (4) the treatment of catch, and (5) the horizon. The two SSB columns are not mixed. The reconstructions are scored separately because they differ in estimated SSB trajectory, assessment formulation, temporal coverage and catch specification; the reference threshold enters only the secondary classification diagnostic and not the primary RMSE comparison. No retention outcome is carried from one specification to the other.

## 2.2 Forecast models

Northern cod surplus production is represented as a discrete-time map, with  $S$  in kt and  $C$  in kt yr<sup>-1</sup>.

**Definition 2.1 (Surplus-production map).** The autonomous surplus-production map with multiplicative depensation is

$$S_{t+1} = [S_t + g(S_t) - C_t + \varepsilon_t]_+, \quad g(S_t) = rS_t(1 - S_t/K) a(S_t),$$

$$a(S_t) = \begin{cases} 1 & (\text{Schaefer; no Allee term declared}), \\ \frac{S_t - \mathfrak{s}}{K - \mathfrak{s}} & (\text{depensation term active, } \mathfrak{s} > 0). \end{cases}$$

Two conventions fix how this map is used. The disturbance  $\varepsilon_t$  is a regression error at estimation only: parameters are fitted to the one-step increments with  $\varepsilon_t$  as the residual, and all forecasts reported here set  $\varepsilon_t = 0$ , so every forecast trajectory is a deterministic plug-in path rather than a multi-step conditional mean, except in M3 and M4, which carry the fitted residual forward as a state. In those two modules the training residuals are  $e_u = \Delta S_u - [g(S_u) - C_u]$  over the training transitions, formed before any clipping. The coefficient  $\hat{\phi}$  is the no-intercept lag-one least-squares estimate  $\sum_u e_u e_{u-1} / \sum_u e_{u-1}^2$ ; it is set to zero when the denominator is not positive or the window supplies fewer than four residuals, and is clipped to  $[-0.95, 0.95]$ . The residual carried into the forecast is the last training residual  $e_{\text{last}}$ , and the projected residual at step  $k \geq 1$  is  $\hat{\phi}^k e_{\text{last}}$ , added inside the update before the state is clipped to  $[10^{-3}, 10^6]$  kt. Catch is dated so that  $C_t$  is the catch taken during the interval from  $t$  to  $t + 1$ ; a forecast issued at origin  $t$  therefore uses the recorded catch for the year it is forecasting. Under the coarse regime this places the 1992 drop in the step from 1992 to 1993, so on the fixed collapse test window (1991–1995) the drop does not enter the first two forecast steps; the models are already rising before it applies. For multi-step forecasts the catch path is supplied for the whole horizon, so the exercise is a conditional hindcast with respect to catch, not an operational forecast; Section 4 returns to this.

The factor  $a(S_t)$  is a multiplicative depensation term, not a parameter switch. Setting  $\mathfrak{s} = 0$  gives  $a(S_t) = S_t/K$  (a cubic modification of the logistic surplus; Pella and Romano, J.P. and Wolf, M., 2005. Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4), 1237–1282. doi:10.1111/j.1468-0262.2005.00615.x.

Tomlinson, 1969), not the Schaefer (1954) law. The Schaefer model is the separate member of the family in which the factor is replaced by 1; Abaee (2026, Periodic review as sampled governance) makes the same distinction.

**Shared symbols across the companion papers.** The companion papers share this vocabulary with paper-local meanings; the letters below carry meanings here that differ there:

| Here                               | In the companions                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $K$ : the carrying capacity        | the safe set and kernel $K$ (obstruction calculus); the full-information kernel (certification); the window depth and contact set $\mathcal{K}(B)$ (minimax duals) |
| $r$ : the intrinsic growth rate    | the catch share $r = C/B_t$ (applied regime viability)                                                                                                             |
| $\sigma$ : the process/noise scale | the kernel time variable $\sigma$ (certification; minimax duals); the floored exit count $\sigma^*$ (worked systems)                                               |

Section SI-2 gives the statement and proof.

The model ladder is given in Table 2.

The ladder is the forward-ordered set of seven entries {persist, mean, M1, M1b, M2, M3, M4} of Table 2, ordered by structural complexity (autonomous surplus map, Allee extension, stock-flow, residual, delay): two naive baselines (persist, mean) and five structural modules.

Where this article refers to a five-module ladder evaluated against two naive baselines, it means this set. It is not a strict nesting for M2 and M4.

**Table 2.** Model ladder.

| ID      | Class                 | Free on the training window                                                   | Frozen into the test window              |
|---------|-----------------------|-------------------------------------------------------------------------------|------------------------------------------|
| persist | baseline              | —                                                                             | $\hat{S}_{t+h} = S_t$                    |
| mean    | baseline              | training mean                                                                 | $\hat{S}_{t+h} = \bar{S}_{\text{train}}$ |
| M1      | autonomous            | $r, K; C = \text{training-mean catch (plugged, not estimated)}$               | same $C$                                 |
| M1b     | autonomous with Allee | $r, K, \mathfrak{s}; C = \text{training-mean catch (plugged, not estimated)}$ | same $C$                                 |
| M2      | stock-flow            | $r, K; C_t$ prescribed                                                        | prescribed $C_t$                         |
| M3      | residual              | M2 + AR(1) residual                                                           | $\phi$ persisted                         |
| M4      | lagged initialisation | $r, K$ and $\phi$ refitted exactly as for M3                                  | forecast starts from $S_{t-1}$           |

The coarse catch regime, taken from DFO (2016) prose (the precautionary-approach framework of DFO, 2009), is  $C_t = 240$  kt for  $t \leq 1991$ ,  $C_t = 120$  kt for  $t = 1992$ , and  $C_t = 5$  kt for  $t \geq 1993$ . Year-by-year landings (Schijns et al., 2021, Table 1) replace that regime in Section 3.2. Regular et al. (2025) Table 1 matches Schijns exactly on 1983–1993 (11 years, maximum absolute difference 0 t); STATLANT matches Schijns on 1983–1993, so the STATLANT-versus-Schijns sensitivity is closed on 1983–1993 (they are the same column there). The Specification A collapse test window runs to 1995, two years beyond the verified match; the residual exposure is immaterial, since Schijns records 1.31 kt in 1994 and 0.41 kt in 1995 against a collapse-window  $\Delta S$  of hundreds of kilotonnes. A 1956 discrepancy between those sources (236,210 t versus 263,210 t) lies outside the 1983–2015 Specification A range; Specification B begins in 1954 and so contains 1956, but it draws its catch column from the xteNCAM reconstruction rather than from either of these two sources, so the discrepancy is unused on both specifications. The coarseness of the regime series is a limitation.

Parameters are estimated by one-step least squares on the training window only. Rolling-origin training windows expand rather than slide: each origin refits on all history up to that origin, subject to the minimum length. A window of  $m$  annual states supplies  $m - 1$  one-step transitions, and it is the transition count that sets the effective regression sample. Bounds:  $r \in (0.001, 2]$ , and  $K$  constrained above the largest state the one-step regression conditions on. The estimation code optimises  $K$  on  $[\max_{\text{train}} S + 10, 5000]$  kt, where  $\max_{\text{train}} S$  is taken over the predictor states  $S_t$  of the training transitions and so excludes the terminal state of the window. Here 500 kt is the multi-start initialiser rather than the lower bound, consistent with the frozen specification’s constraint of  $K$  above the training maximum. The remaining estimation bounds are  $\mathfrak{s} \in [0, \max_{\text{train}} S_t]$  with the further feasibility restriction  $0 < \mathfrak{s} < 0.8K$  enforced in the objective,  $\phi$  clipped to  $[-0.95, 0.95]$ , and the index exponent  $b$  fitted jointly with  $r$  and  $K$  without bounds. The profile of Section 3.2 restricts  $\mathfrak{s}$  further, to  $[0, \min_{\text{train}} S_t]$ , for the reason given there. The reported fits attain the upper endpoint ( $K = 5000$  kt) where the objective is flat out to the imposed bound; M1b’s recovery-window  $K = 105.8$  kt is a valid interior fit, not a bound violation. Where a reported fit sits at 500.0 kt this is the multi-start initialiser, which the optimiser has not left because the objective is flat there, and not an active constraint. M3’s autoregressive coefficient is estimated in a second stage:  $r$  and  $K$  are fitted first by one-step least squares, and  $\phi$  is then obtained as the lag-one regression coefficient of the resulting training residuals on their own lag, clipped to  $[-0.95, 0.95]$ ; it is not jointly optimised with  $r$  and  $K$ . Every fitted-parameter value this article prints, with the window it belongs to

and the bound it attains, is collected in Table 10; the per-origin rolling fits and the parameters the article does not print are archive items, marked there as such.

A survey-start variant replaces the surplus initial condition by  $\hat{q} I_t$ , where  $I_t$  is the autumn research-vessel abundance index and  $\hat{q}$  is the training-window median of  $\text{SSB}/I$ . The index is an NCAM input, not an independent stock.

Prey-informed variants (Section 3.4) scale surplus production by a 1991 capelin regime or by the tabulated 3L spring acoustic index (DFO, 2024b). Pre-1991 acoustic values are not carried across 1991. Missing survey years are not interpolated; within the post-1991 regime the last eligible observation is carried forward, and training transitions with no eligible index value are dropped from the fit.

### 2.3 Evaluation design

**Table 2b.** Information set at forecast origin  $t$ . “Available” means the quantity is dated at or before  $t$  and is used by the module named; “supplied” means the quantity is dated after  $t$  and is nevertheless provided to the forecast, which is what makes the exercise a conditional hindcast rather than an operational test.

| Quantity                          | Status at origin $t$                                                    | Used by                                      |
|-----------------------------------|-------------------------------------------------------------------------|----------------------------------------------|
| Target series $S_u$ , $u \leq t$  | available; single retrospective vintage, not the vintage current at $t$ | all modules and baselines                    |
| Origin state $S_t$                | available                                                               | all except M4                                |
| Lagged state $S_{t-1}$            | available                                                               | M4 (forecast start)                          |
| Training transitions $u \leq t$   | available; window expands with $t$                                      | parameter estimation, all structural modules |
| Fitted residual $e_{\text{last}}$ | available; last training transition                                     | M3, M4                                       |
| Catch $C_u$ , $u \leq t$          | available                                                               | M1, M1b (training mean); M2–M4               |
| Catch $C_u$ , $u > t$             | <b>supplied</b> along the horizon                                       | M2, M3, M4                                   |
| Prey index $I_u$ , $u \leq t$     | available; last observation carried forward                             | M_cap_index                                  |
| Prey index $I_u$ , $u > t$        | not used                                                                | —                                            |
| Reference point (LRP)             | fixed by specification                                                  | secondary Brier score only                   |

The distinction the table draws — between quantities dated at or before the origin and quantities dated after it but supplied anyway — is not specific to this study, and reporting it explicitly is proposed here as a template for hindcast evaluations generally: an evaluation that does not state which of its inputs were unavailable in real time cannot be read as evidence about operational skill. The single row that makes this a conditional hindcast is the supplied future catch: M2, M3 and M4 receive the realised catch path over the forecast horizon, which no operational forecast would have. That asymmetry strengthens rather than weakens the comparison: the three modules given information no operational forecast could possess still do not attain lower pooled rolling RMSE than a rule that uses only the current state. Persistence receives no catch information at all. M1 and M1b do not, using instead the training-mean catch, which is why their scores are nearly invariant to the catch treatment. The target series is a single retrospective reconstruction throughout, so no module sees the assessment vintage that would have been current at its origin.

**Table 2c.** Rolling-origin sets, as constructed by the scored code and archived per experiment. Minimum training length is eight years for the naive baselines throughout, eight for the Specification A ladder, and twelve for the Specification B ladder; the baseline and ladder origin sets on Specification B therefore differ, and every comparison reported against a baseline is recomputed on the module’s own origins.

| Experiment                    | $h$ | $n$ | Origin range |
|-------------------------------|-----|-----|--------------|
| Specification A ladder        | 1   | 25  | 1990–2014    |
| Specification A ladder        | 5   | 21  | 1990–2010    |
| Specification B ladder        | 1   | 59  | 1965–2023    |
| Specification B ladder        | 5   | 55  | 1965–2019    |
| Specification B baselines     | 1   | 63  | 1961–2023    |
| Specification B baselines     | 5   | 59  | 1961–2019    |
| M_cap_index, Specification A  | 1   | 24  | 1991–2014    |
| M_cap_index, Specification A  | 5   | 20  | 1991–2010    |
| M_cap_index, Specification B  | 1   | 36  | 1988–2023    |
| M_cap_index, Specification B  | 5   | 32  | 1988–2019    |
| M_cap_regime, Specification A | 1   | 25  | 1990–2014    |
| M_cap_regime, Specification B | 1   | 59  | 1965–2023    |

The index experiments start later than the ladder because an origin is scored only where a carried-forward index value exists. On Specification B that admits three pre-break origins (1988–1990), which draw on pre-1991 surveys; the carry is reset at 1991 so that no pre-break value is carried into the post-break regime.

What the design does and does not test should be stated before the scores. The predictand at every origin is a single retrospectively reconstructed assessment series, not the assessment vintage that would have been available at that origin; assessment vintages were not available for this evaluation. Earlier SSB values therefore embed information produced after the nominal forecast date, and the smoothing inherent in an assessment reconstruction is present in both the target and the persistence baseline. Catch is supplied along the forecast horizon from the catch file rather than forecast. The exercise is accordingly a retrospective prediction experiment and, with respect to catch and the prey index, a conditional hindcast; it is not a test of operational forecasting skill, and the comparison against persistence inherits the same status. Section 4 returns to what this does and does not license.

Fixed windows on Specification A: collapse, train 1983–1990, test 1991–1995; recovery, train 1995–2007, test 2008–2015. The year 1995 appears in the collapse test window and again as the first recovery training year. The two fixed-window exercises are separate fits scored separately and neither informs the other, so this is not leakage within an exercise; it is noted because the two windows are reported side by side. Rolling origin: minimum eight training years; horizons  $h = 1$  and  $h = 5$ . Estimation is by one-step least squares on the training window, while the five-year score evaluates iterated trajectories of parameters fitted for one-step fit — a training/testing cost-function mismatch; trajectory-matched estimation over the multi-year horizon is an untested alternative, not scored here.

The primary score is RMSE of SSB (kt), taken over all scored origin–target pairs at the stated horizon; the  $h = 5$  score is the root mean square of the five-year-ahead errors themselves, not an average along the intervening trajectory. Log-RMSE uses natural logarithms of states floored at  $10^{-3}$  kt. The sign-hit rate compares  $\text{sign}(\Delta S)$  between forecast and observation over adjacent years within a window, giving one fewer comparison than there are scored years, and is reported as NA where a rule supplies no directional prediction rather than as a zero hit rate. Secondary scores are mean absolute error, RMSE on  $\log S$ , the threshold misclassification rate for  $\mathbf{1}\{\hat{S} < \text{LRP}\}$  — equivalently the Brier score for a deterministic binary forecast, which is what a hard 0/1 prediction makes it, so no probability forecast is implied — and the sign-hit rate of  $\Delta S$  on fixed windows. On Specification A the Brier indicator rests on a degenerate event — the 884.6-kt LRP sits above the entire origin range, so the indicator is 1 at every origin and persistence attains 0.00 by construction. The score does separate models in this sample (persistence and the training mean at 0.00 against 0.80–1.00 for the structural modules, Table 3), but it separates them on an outcome that never varies, so it carries no early-warning information and is reported as a secondary diagnostic only (Section 3.3 gives the record). The sign-hit rate is

conventionally 0.00 for persistence, whose forecast  $\Delta S = 0$  has no sign; persistence is excluded from that ranking by declaration.

A descriptive scale comparison is available because the extended assessment publishes an uncertainty band on its own predictand: `xtencam_table17_ssb.csv` carries 95% limits for every year, with a median relative width of 0.44. Writing  $D = |\hat{S}_{t+h|t} - S_{t+h}|/[(\text{hi}_{t+h} - \text{lo}_{t+h})/2]$ , the proportion of hindcast errors smaller than half the published interval width is, on Specification B, 56% for persistence, 36% for M1 and 59% for M3 at  $h = 1$  ( $n = 59$ ), and 18%, 5% and 7% at  $h = 5$  ( $n = 55$ ).

These are **not** coverage probabilities and are not comparable to a nominal 95%. The published interval expresses uncertainty about true biomass given the assessment's data and model; re-centring its width on a hindcast produces an interval with no established coverage, and the bounds are in any case right-skewed on the biomass scale (67 of 71 years asymmetric by more than 10%, close to symmetric only in logarithms), so a symmetric normal reading of the width would be the wrong transform. Nor does the statistic establish that model *rankings* are robust to assessment uncertainty: it compares each model's errors with a target-state interval, not the differences between models. It indicates only that hindcast discrepancies are mostly of the same order as, or larger than, reported target-state uncertainty. On this statistic M3 (59%) reads marginally above persistence (56%) at  $h = 1$ , which the rolling-origin RMSE ordering does not; the two need not agree, and no retention verdict rests on either this statistic or that reading.

The retention rule is stated as explicit hypotheses.

**Retention rule.** *A module  $M$  on the ladder of Table 2 is retained only if all of the following hold on the rolling-origin primary RMSE score:*

- (H1)  *$M$  reduces primary RMSE relative to its declared comparator — the next-simpler rung by complexity ordering (M1b against M1, M3 against M2). Only M3 against M2 is a strict nesting: M1b is a separate branch of Definition 2.1 rather than a parameter restriction of M1, since  $s \rightarrow 0$  yields the cubic factor  $a(S) = S/K$  and not the Schaefer law (Section SI-2), so its comparator is declared by ordering rather than by nesting. For the two rungs that are not nestings at all, M1 for M2 (the autonomous constant-catch map whose catch treatment M2 changes) and M3 for M4 (the module the lagged initialisation acts on). M1b is reported as the alternative comparator for M2 and never decides retention. M1 is the first structural rung and has no simpler structural comparator, so H1 is vacuous for M1 and its retention turns on H2 and H3;*
- (H2)  *$M$  reduces primary RMSE relative to last-value persistence;*
- (H3) *each required reduction holds at both horizons  $h = 1$  and  $h = 5$  — the frozen specification's primary score is the rolling-origin RMSE pair — and exceeds a tie band of 5% of the comparator's score; improvements inside the band are ties and do not retain.*

Retention rule (as scored; band = 0.05, horizons = {1, 5})

```
for each module M on the ladder:
  retain <- TRUE
  for h in horizons:
    if RMSE(M, h) >= (1 - band) * RMSE(persistence, h):
      retain <- FALSE                                # H2: beat persistence
    if comparator(M) exists and
      RMSE(M, h) >= (1 - band) * RMSE(comparator(M), h):
      retain <- FALSE                                # H1: beat next-simpler
  report retain                                       # H3: both horizons, band
```

Scores are origin-matched: every comparison uses the module's own origin set, and the baseline is recomputed on that set rather than taken from a longer one. The rule is deliberately

conservative in two places, and Section 3.7 quantifies what each costs: the band suppresses retention on margins inside sampling noise, and the comparator requirement forbids a module from being retained merely because it beats persistence when a simpler member of the same ladder does so too.

*Retention is decided separately on each specification. A module failing any of (H1)–(H3) is not retained. Two disclosures complete the rule. First, it is a point-rule ranking under a pre-set score, not a test of equal predictive ability; the uncertainty analysis of Section 3.5 attaches Diebold–Mariano and moving-block-bootstrap intervals to its margins and changes no verdict. Second, the tie band and the comparator declarations are completions recorded after the scores of Section 3 were computed: no recorded verdict depends on them — the smallest structural deficit against persistence (M1b’s on Specification A at  $h = 1$ ;  $(114.80 - 98.05)/98.05 = 17.1\%$ ) is far outside the 5% band, and no (H1) comparison in Tables 3–8 reverses under either comparator reading.*

A module is not retained when the one-step least-squares implementation evaluated here fails at least one of (H1)–(H3) on the rolling-origin primary RMSE score on the series tested. Non-retention is a statement about these implementations on these series, not about every possible implementation of the underlying model class, and it is weaker than a statistical null result: it records that a module did not clear a stated bar, not that no difference exists.

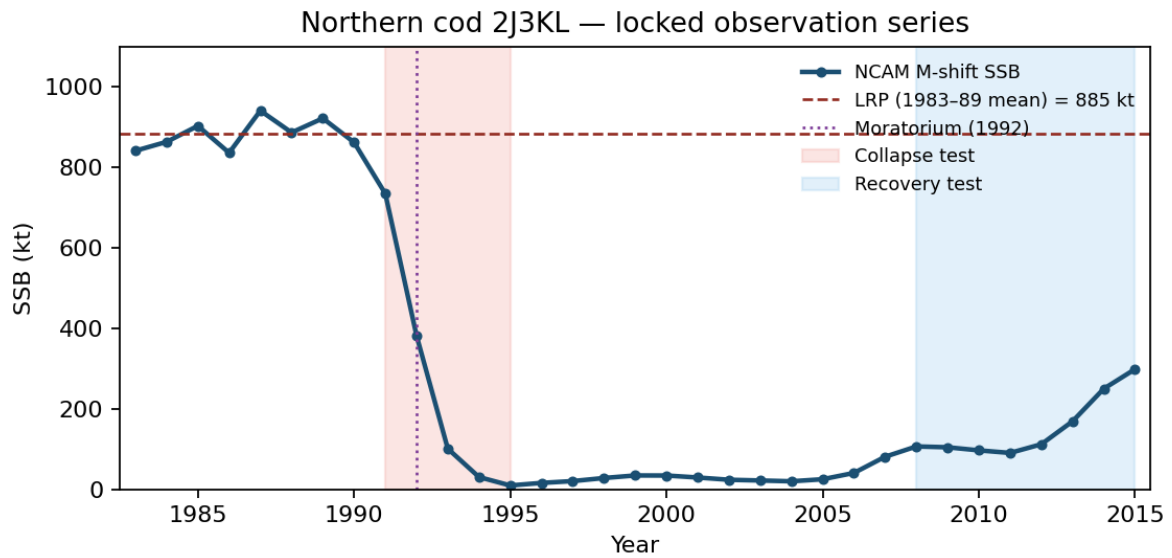
The scoring is deterministic: the scripts are listed in the Data availability statement with pinned output checksums and an archived environment specification, so the reported arithmetic regenerates byte for byte within that environment. Cross-environment numerical identity is not claimed, and one M1b row is environment-sensitive at the  $\pm 17$ -kt level, as recorded there. Reproducibility of the arithmetic is distinct from the analytical result of Section 4, which is proved rather than computed.

On Specification B the collapse window is train 1954–1989, test 1990–1995, and the recovery-stall window is train 1995–2012, test 2013–2024. As on Specification A, 1995 appears both in the collapse test window and at the start of the later training window; the two windows are scored independently and no fitted quantity crosses between them, so the shared year is not leakage. Catch is Regular et al. (2025) Table 1 landings (2024 persisted from 2023). No row is taken from NCAM 2016. The ladder’s rolling origin on Specification B uses a minimum of twelve training years ( $n = 59$  at  $h = 1$ ,  $n = 55$  at  $h = 5$ ); the naive baselines retain the eight-year minimum ( $n = 63$  and  $n = 59$ ). The two origin sets therefore overlap but are not identical, so a baseline computed over the longer naive origin sets would mix four extra early origins into the comparison. Table 6 accordingly reports the origin-matched baselines, computed on the ladder’s own origins: on the identical twelve-year origins persistence reads 84 kt ( $n = 59$ ) at  $h = 1$  and 300 kt ( $n = 55$ ) at  $h = 5$ , and the training mean 458 kt and 522 kt, against M1’s 120 kt and 432 kt. Persistence remains the lowest-RMSE forecast. The mixed-origin reading (88 kt, quoted in the Table 6 caption) exceeds the controlled reading of 84.4 kt by 3.6 kt — an origin-mix effect of no retention consequence. On the identical post-break origins of the Table 7 control ( $n = 33$ ,  $h = 1$ ) the recomputed baseline is 75 kt, matching the post-break value reported with the two-regime control of Section 3.4. On the index modules’ origin sets (Table 8), persistence reads 97 kt ( $n = 24$ ,  $h = 1$ ) and 193 kt ( $n = 20$ ,  $h = 5$ ) on Specification A, and 79 kt ( $n = 36$ ) and 288 kt ( $n = 32$ ) on Specification B. The index module loses to the origin-matched baseline in every cell, and the verdict — the module is not retained — is unchanged.

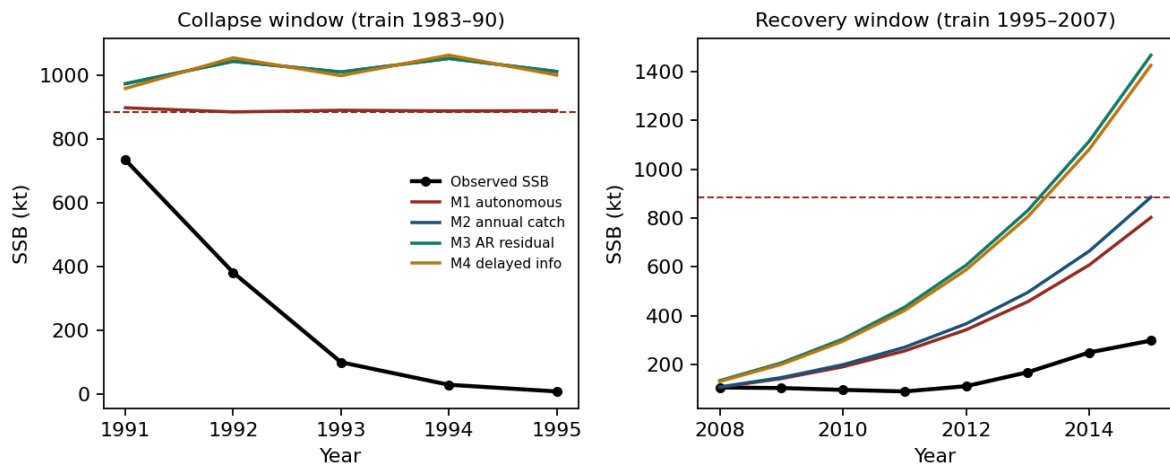
### 3. Results

#### 3.1 Primary specification

**Table 3.** Fixed-window scores (RMSE in kt), with both baselines reported on every window, computed on the same fixed origins set in Section 2.3. Table 3 is not to be read as a retention table — retention is decided on rolling-origin primary RMSE only.



**Figure 1:** NCAM M-shift SSB (DFO, 2016, Table A2). Dashed line: 2016 LRP = 884.6 kt. 2015 SSB is 33.8% of that LRP, matching the advisory report statement of 34%.



**Figure 2:** Multi-step forecasts from the end of each training window. Recovery is under-predicted when an AR residual fitted on a slow early-recovery train is persisted.

| Window   | Model   | RMSE      | MAE | log-RMSE | Brier | Direction |
|----------|---------|-----------|-----|----------|-------|-----------|
| Collapse | persist | 670       | 610 | 2.71     | 0.00  | NA        |
|          | M1      | 694       | 638 | 2.73     | 1.00  | 0.50      |
|          | M1b     | 694       | 636 | 2.73     | 0.80  | 0.00      |
|          | M2      | 819       | 751 | 2.85     | 1.00  | 0.25      |
|          | M3      | 819       | 751 | 2.85     | 1.00  | 0.25      |
|          | M4      | 819       | 750 | 2.85     | 1.00  | 0.25      |
|          | mean    | 688       | 630 | 2.72     | 0.00  | NA        |
| Recovery | persist | 104       | 72  | 0.69     | 0.00  | NA        |
|          | M1 = M2 | 120       | 105 | 0.61     | 0.00  | 0.57      |
|          | M1b     | <b>90</b> | 55  | 0.52     | 0.00  | 0.57      |
|          | M3      | 220       | 200 | 0.92     | 0.00  | 0.57      |
|          | M4      | 214       | 195 | 0.91     | 0.00  | 0.57      |
|          | mean    | 144       | 124 | 1.60     | 0.00  | NA        |

On collapse, fitted  $r$  reaches 1.935, close to but not at the imposed upper bound of 2. The 1983–1990 window is a high, weakly trending stock. The fitted pre-collapse model does not reproduce the subsequent decline, and dropping  $C$  from 240 to 5 raises forecast SSB. In the model the 1992 drop is supplied as an exogenous catch path; historically it was largely a policy response to depletion. Because production is stationary, prescribing the lower catch in a constant- $r$  stock-flow equation raises the predicted increment and forces a mechanical rebound instead of reproducing the crash. Within this accounting, had the crash been driven by the catch regime one would expect supplying the catch path to help; M2 does not improve on M1. Supplying the prescribed catch path did not improve this implementation’s collapse-window hindcast. That is not proof of one alternative: a misspecified and weakly identified module need not improve even when the variable it adds is genuinely relevant, so the result bounds what this ladder can attribute rather than identifying the mechanism.

Under the coarse catch regime, M1 and M2 coincide on the recovery window; under annual landings they do not (264 against 303 kt on the fixed recovery window, Section 3.2). The reason is structural:

On the recovery window (train 1995–2007, test 2008–2015) of Specification A, M1 and M2 produce identical forecasts: M1 plugs the training-mean catch into the autonomous map and M2 prescribes  $C_t$  year by year, but  $C_t \equiv 5$  kt on both train and test windows, so the two prescriptions coincide and the least-squares  $(r, K)$  optimum is identical.

M1b reports a lower RMSE (90 kt against M1’s 120 kt), but  $\mathfrak{s}$  descends towards zero and  $K$  settles just above the training predictor range at 105.8 kt: an unidentified Allee parameter, not a biological threshold. The estimate approaches the lower boundary to numerical tolerance, for a reason visible in the objective.

**Observation 3.2 (Boundary Allee estimate on the recovery window).** Any  $\mathfrak{s}$  strictly above the training minimum makes the depensation factor  $a(S_t) = (S_t - \mathfrak{s})/(K - \mathfrak{s})$  negative for observations below  $\mathfrak{s}$ , so wherever the observed  $\Delta S$  is positive at such a state the fit incurs a large squared residual. On the 1995–2007 recovery window seven of the twelve annual increments are positive — the five negative ones fall in a single consecutive run, 1999–2004 — and on balance this penalty pushes the estimate downward: the fitted  $\mathfrak{s}$  descends towards zero ( $2.1 \times 10^{-23}$  under the coarse regime and  $9.4 \times 10^{-6}$  under annual landings) while  $r$  reaches its upper bound of 2 and  $K$  settles just above the training range at 105.8 kt. Zero is excluded by the objective’s feasibility restriction  $0 < \mathfrak{s} < 0.8K$ , so these values are a numerical approach to an unattained infimum rather than an attained boundary estimate. At such values the fitted object is effectively the zero-threshold cubic branch  $a(S) = S/K$  of Definition 2.1, not a positive-threshold Allee model, so M1b’s lower error here is not evidence for depensation:  $r$  and  $K$  absorb what  $\mathfrak{s}$  does not.

This is an empirical finding about these fits, not a theorem. Two qualifications matter. The pressure just described is a tendency and not a proof that the global optimum must lie below

the training minimum: the objective trades errors across all observations, and  $r$ ,  $K$  and  $\mathfrak{s}$  can compensate (for M1b the catch is plugged, not estimated, and the profile re-optimises only  $r$  and  $K$ ) for one another. The window is also not monotone — SSB declines in five consecutive annual decreases, from 34.59 kt in 1999 to 20.07 kt in 2004, before rising to 81.10 kt in 2007 — so no argument that assumes a monotone recovery applies to it. Nor is a boundary estimate by itself proof of non-identifiability, and a biologically meaningful threshold may lie outside the sampled range.

A profile of the least-squares objective shows how weakly the data constrain  $\mathfrak{s}$ . The grid for  $\mathfrak{s}$  spans the profile range  $[0, \min_{\text{train}} S_t] = [0, 9.68]$  kt. Two different ranges are in play and should not be conflated. Both are built from the *predictor* states of the training window — the  $S_t$  that the one-step regression conditions on, which exclude the terminal response — so on the recovery window (states 1995–2007) the relevant maximum is  $S_{\text{max}}^{\text{pred}} = 40.83$  kt over the twelve transitions beginning 1995–2006, not the 81.10 kt terminal state of 2007.

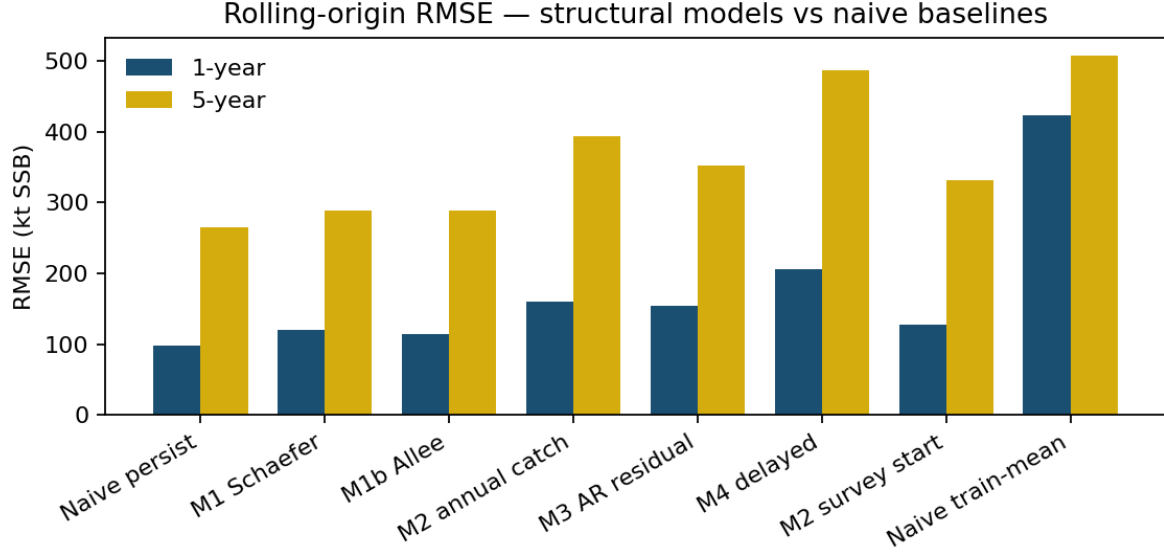
| Quantity                    | Range                                              | Recovery window    |
|-----------------------------|----------------------------------------------------|--------------------|
| $K$                         | $[S_{\text{max}}^{\text{pred}} + 10, 5000]$ kt     | $[50.83, 5000]$ kt |
| $\mathfrak{s}$ , scored fit | $(0, S_{\text{max}}^{\text{pred}}]$ , open at zero | $(0, 40.83]$ kt    |
| $\mathfrak{s}$ , profile    | $[0, \min_{\text{train}} S_t]$                     | $[0, 9.68]$ kt     |

The profile is computed on the narrower range so that the production multiplier is non-negative at every predictor state. That is an interpretive restriction adopted for the profile, not a feasibility condition of the model and not a statistical ground for excluding higher thresholds: a negative multiplier can be compatible with negative observed increments, and the objective trades errors across observations. Both ranges therefore exclude a threshold placed above the collapsed range: at  $\mathfrak{s} = 50$  kt — outside the scored bound — the factor  $a(S)$  would be negative at all twelve predictor states, so the model would predict negative production across the whole window. The estimate at the lower edge is consequently a statement about a restricted feasible set, and this design neither supports nor excludes depensation hypotheses — a predator pit, for instance — that place the stock below a high threshold throughout the recovery. Testing one requires lifting the range bound, which is outside the frozen specification scored here. Fixing  $\mathfrak{s}$  on that grid and re-optimising  $r$  and  $K$  at each value gives a profile objective that rises only from 126.50 to 135.04 kt<sup>2</sup> under the coarse regime, and from 122.90 to 126.50 kt<sup>2</sup> under annual landings. Expressed as  $n_{\text{tr}} \log\{\text{SSE}(\mathfrak{s})/\text{SSE}(\hat{\mathfrak{s}})\}$  with  $n_{\text{tr}} = 12$ , the largest deviation across the grid is 0.78 under the coarse regime and 0.35 under annual landings. These are descriptive measures of curvature, not calibrated likelihood-ratio tests: the least-squares fit carries no error model, the estimate  $\hat{\mathfrak{s}}$  lies on the boundary of its range,  $r$  rests at its own upper bound, the window supplies twelve transitions, and the states are assessment reconstructions rather than direct observations. Any one of these would compromise a  $\chi^2_1$  reference. The substantive reading does not depend on the reference distribution: the objective is nearly flat across the whole range, so the data do not locate  $\mathfrak{s}$  within it.

The mechanism is parameter compensation. In  $a(S_t) = (S_t - \mathfrak{s})/(K - \mathfrak{s})$ , raising  $\mathfrak{s}$  reduces the numerator while lowering  $K$  raises  $1/(K - \mathfrak{s})$ , and along the profile  $K$  slides from 106.0 to 82.7 kt (coarse) or 129.8 to 95.6 kt (annual landings) while  $r$  remains at its upper bound of 2 throughout. The product  $rS_t(1 - S_t/K)a(S_t)$  is nearly invariant along this path, so the data cannot separate  $\mathfrak{s}$  from  $K$ . The training observations therefore do not constrain behaviour near a putative threshold, and the boundary estimate is not evidence about a biological Allee threshold in either direction.

M3's  $\phi = 0.95$  persists a negative residual, which raises its error relative to persistence on this window; against its own declared comparator M2 it still reads lower.

**Table 4.** Rolling-origin summary, Specification A, coarse catch regime.



**Figure 3:** Rolling-origin RMSE. Persistence is the best one-year and five-year point forecast.

| Model   | $h = 1$ RMSE | $h = 1$ MAE | $h = 1$ log-RMSE | $h = 5$ RMSE |
|---------|--------------|-------------|------------------|--------------|
| persist | <b>98</b>    | <b>48</b>   | <b>0.52</b>      | <b>265</b>   |
| M1      | 121          | 80          | 8.02             | 289          |
| M1b     | 115          | 80          | 8.70             | 289          |
| M2      | 144          | 61          | 0.59             | 398          |
| M3      | 135          | 53          | 3.39             | 366          |
| M4      | 196          | 82          | 0.76             | 488          |
| mean    | 424          | 375         | 2.35             | 507          |

Log-RMSE for M1 and M1b is large because some origins produce trajectories that hit the numerical floor. M2 keeps the state off zero and stabilizes the log score, but loses on raw RMSE. M4, the only model started from a year-old state, is the worst structural model on raw RMSE; its decomposition against a stale-start persistence control is reported in Section 4.

No structural model — M1 through M4 — is retained on Specification A. Persistence is the lowest-RMSE forecast.

### 3.2 Annual landings and survey start

Year-by-year landings for 2015 are 4.436 kt, matching DFO reported landings. Pre-collapse catches are 172–269 kt, not a flat 240. The 1992 drop is to 41 kt, then 11 kt, then 0.4–1.9 kt.

**Table 5.** Rolling-origin RMSE (kt) with annual landings.

| Model            | $h = 1$   | $h = 5$    |
|------------------|-----------|------------|
| persist          | <b>98</b> | <b>265</b> |
| M1               | 121       | 289        |
| M1b              | 115       | 289        |
| M2               | 160       | 394        |
| M3               | 154       | 352        |
| M4               | 206       | 486        |
| M2, survey start | 128       | 331        |

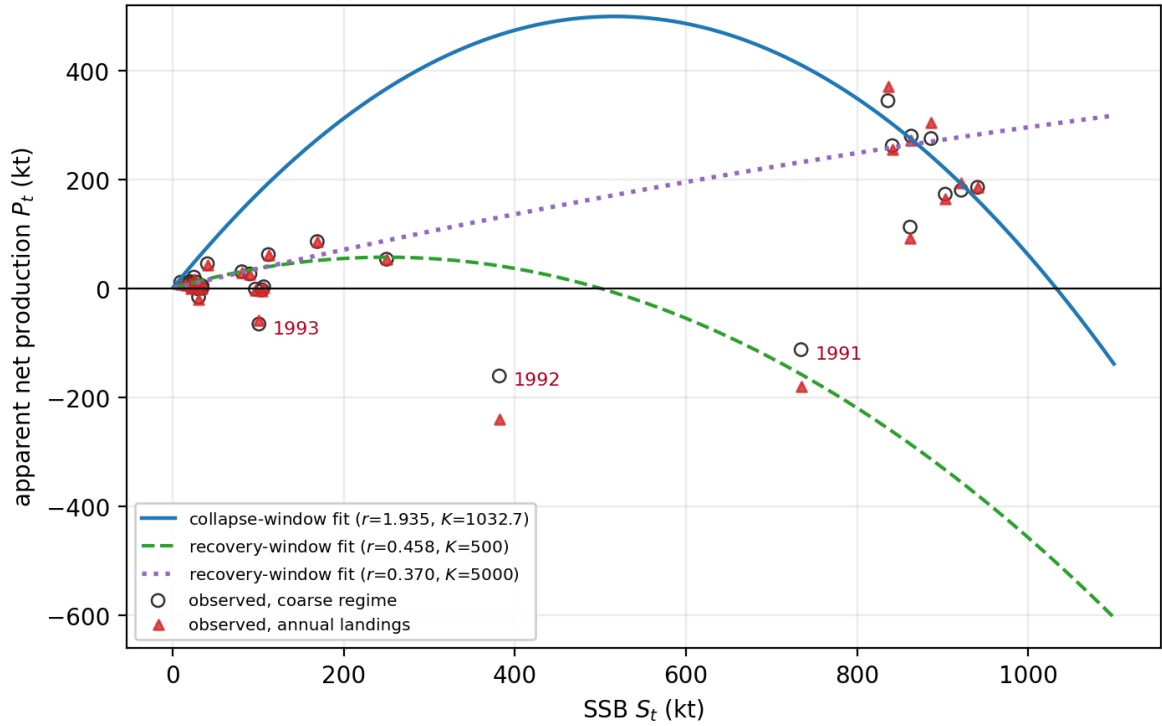
Annual landings make M2 worse than the coarse regime on one-year RMSE (160 versus 144 kt). Collapse-window RMSE under annual landings remains about 821 kt for M2, M3 and M4

(819 kt under the coarse regime; M1 and M1b are at 694 kt on both treatments, Table 3). On the annual-catch recovery window (train 1995–2007, test 2008–2015) the ladder scores M1 264, M1b 78, M2 303, M3 609, and M4 586 kt, against a frozen-persistence error of 104 kt (forecast held at the 2007 level of 81 kt). Annual landings therefore do not repair the recovery-window fit, and the residual-persistence models deteriorate severely (609 and 586 kt against 220 and 214 kt under the coarse regime). M1b’s 78 kt carries the same unidentified-Allee caveat as its regime-window fit.

Apparent net production  $S_{t+1} - S_t + C_t$  is strongly negative in 1991–93 even after subtracting reconstructed catch. This quantity is a diagnostic construct, not an exact stock balance:  $S$  here is spawning-stock biomass while  $C_t$  is total landings, and changes in SSB also reflect maturation into the spawning stock, changing maturity and weight-at-age schedules, and the age composition of catch and of natural deaths. Within the surplus-production accounting the ladder itself assumes — in which  $S$  is the modelled biomass state and  $C_t$  is subtracted from it — matching the observed  $\Delta S$  with no production at all would require removals of order  $-\Delta S$ , and the observed decline is far larger than  $C_t$ . On that reading the unexplained term dwarfs the difference between the two catch treatments. Regular et al.’s  $M \approx 2.5$  peak is an age-structured assessment’s estimate informed by additional data and assumptions; it is not the same object as this scalar residual, though the two point in the same direction. Within the model class tested, substituting annual recorded landings for the coarse regime does not produce the crash in a constant- $r$  surplus model.

Reading the same quantity as an implied productivity makes the non-stationarity explicit. Inverting the Schaefer form at a fixed  $K$  gives  $\hat{r}_t = P_t/[S_t(1 - S_t/K)]$ , a reconstruction of productivity rather than a forecast model. Periods below are labelled by the year in which a transition starts, so the last usable transition begins in 2014. At  $K = 1032.7$  kt the annual-landings series averages +1.82 over the eight transitions beginning 1983–1990, turns negative through 1991–1994 (mean  $-0.79$ , all four negative), and reads +0.37 over 1995–2007 (thirteen) and +0.25 over 2008–2014 (seven). At  $K = 5000$  kt the sign pattern is the same — +0.32,  $-0.55$ , +0.36, +0.22 — but the magnitudes are not: the post-collapse mean is 0.20 of the pre-collapse mean at the fitted  $K$  and 1.12 of it at  $K = 5000$  kt. The robust feature is therefore the sign reversal, not the size of the pre-to-post contrast, and no constant  $r$  reproduces a sequence that changes sign. The ratio is undefined only at  $S_t = 0$  and  $S_t = K$ ; for  $S_t > K$  it remains defined but its denominator is negative, so the reading as a positive productivity fails. The coarse-regime recovery-window fit, whose  $K$  is 500 kt, cannot be read this way at all over the pre-collapse states, which exceed its  $K$ . The reconstruction above therefore uses the collapse-window  $K = 1032.7$  kt and the annual-landings  $K = 5000$  kt, both of which exceed every observed state.

One numerical disclosure travels with the window: the annual-landings M1 (264 kt) and M1b (78 kt) differ from the coarse-regime recovery rows (M1 = 120, M1b = 90) although both treatments sit on the same SSB column. The reconciliation is that M1’s constant catch is not estimated on that column. It is the training mean of the catch file: 5.0 kt for the coarse regime, and 3.19 kt (the 1995–2007 mean of the annual landings) for the annual treatment. The two least-squares objectives are nearly flat at their minima (MSE 128.35 versus 127.84 kt<sup>2</sup>), so the  $(r, K)$  minimizer slides along the valley as the constant changes:  $r = 0.458$  with  $K$  resting at the multi-start initialiser of 500.0 kt — not a bound, the lower bound on this window being 50.8 kt — against  $r = 0.370$  with  $K$  at its upper bound, 5000.0 kt. The optimiser leaves  $K$  at its starting value in this fit; the profile of Section 3.2, not the optimiser’s behaviour, is what measures the flatness. The same geometry explains the M1b pair ( $r$  pinned at its upper bound in both treatments;  $K = 105.8$  against 129.8 kt): the dependence of the M1b fit on the catch file is the flat-objective identification of a two-parameter autonomous fit on a 13-year window, not an inconsistency between fitting objects. The same flat valley caps what the ranking can resolve. Refitting the recovery-window Schaefer map with  $K$  fixed anywhere on [60, 5000] kt



**Figure 4:** Apparent net production  $P_t = S_{t+1} - S_t + C_t$  against  $S_t$  under both catch treatments, with the fitted collapse-window and the two recovery-window Schaefer curves overlaid.  $P_t$  is a diagnostic construct, not a closed biomass budget: it inherits any error in the catch reconstruction and any assessment revision in the SSB series, and it is not a mass balance. The 1991–1993 points lie below every fitted curve under both treatments, and the two recovery-window parameterisations are almost indistinguishable over the observed range of  $S_t$  while diverging widely when extrapolated beyond it.

and  $r$  re-optimised moves the one-step training objective only from mean squared error 127.4 to 149.9 kt<sup>2</sup> (training RMSE 11.29–12.24 kt, a spread under 0.95 kt), while  $r$  compensates from 0.435 to 0.773. On this window the  $(r, K)$  pair is not identified, and the ordering of valley variants is not a robust ranking.

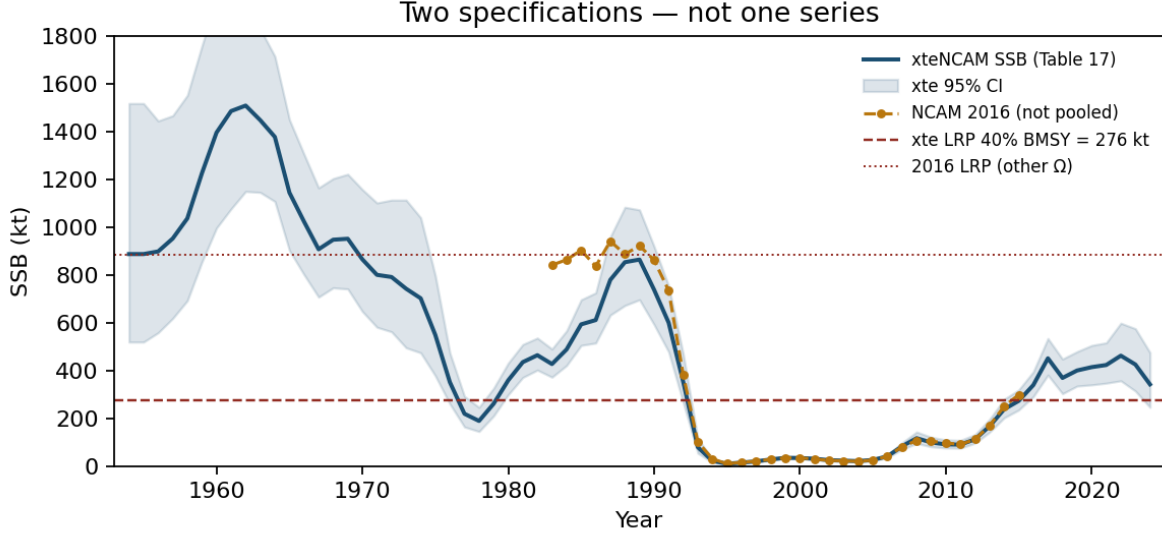
The collapse window is weakly identified in the complementary direction, though less severely than the recovery window. There the stock is high and weakly trending under large catch, so the data approximately impose the single condition  $rS(1 - S/K) \approx C$  on two parameters. Refitting with the estimator of Section 2.3 at fixed  $K$  — the same objective on increments, the same bounds and multi-start — raises the training root mean squared error from 31.3 kt at the fitted  $K = 1032.7$  kt to 56.3 kt at  $K = 1500$  kt and 67.0 kt at  $K = 5000$  kt, while  $r$  falls from 1.935 to 0.674 and 0.331. The objective therefore does discriminate here: a more than twofold rise in training error is not a flat ridge. What the window does not do is pin the pair tightly enough for the implied curve maxima to agree, and the two windows fail in complementary directions — a stock near  $K$  constrains the product more than  $r$ , and low-biomass observations constrain  $K$  only weakly, because the density-dependent term  $\partial g / \partial K = rS^2 / K^2$  is small relative to the approximately linear component rather than absent.

The implied maxima of the fitted curves make the consequence concrete. The formal maximum  $rK/4$  and the reference state  $K/2$  of each printed parameterisation are 500 kt and 516 kt for the collapse window, 57 kt and 250 kt for the coarse-regime recovery fit, and 463 kt and 2500 kt for the annual-landings recovery fit. These are maxima of the fitted scalar curves, not sustainable-yield estimates for this stock: the accounting caveat above — SSB net of catch is not a closed biomass budget — applies to them unchanged. A spread of under 1 kt in recovery-window training RMSE nevertheless separates curves whose formal maxima differ by an order of magnitude. The fitted  $r$  is correspondingly hard to read as a demographic rate: in this discrete annual map the low-biomass multiplier is  $1 + r$ , so  $r = 1.935$  corresponds to an annual factor of 2.94 and a continuous-equivalent  $\log(1 + r) = 1.08 \text{ yr}^{-1}$ , which is high for a late-maturing gadoid but is not the same quantity as a continuous-time intrinsic rate of increase, and SSB can also change through maturation of existing cohorts. These are identification limits with biological content, not numerical curiosities. One mechanism unifies both faces of the catch-dependence: a constant catch shifts the flat valley’s minimiser without moving the objective, so the rolling M1 and M1b scores (refit at every origin) are insensitive to the catch treatment — both are identical after the displayed rounding across the two treatments (the archived values differ by at most 0.04 kt), because future catch is not supplied to these modules — though the training-mean catch does change with treatment and can move their fitted parameters, leaving the rolling scores at 121 and 115 kt at  $h = 1$  — while the single fixed-window fit is not. The property is specific to the two constant-catch modules: M2, M3 and M4 take  $C_t$  into the state update at every step, and their one-year rolling scores do move with the treatment (144 to 160, 135 to 154, and 196 to 206 kt respectively).

Starting from  $\hat{q}I_t$  instead of SSB gives one-year RMSE 128 kt (still worse than persist 98); one-year log-RMSE 0.49 versus persist 0.52; five-year RMSE 331 versus persist 265. On the primary score the variant is not retained. The log-score difference is recorded and not used for selection. The survey index contributes to NCAM and therefore does not provide an independent validation target.

### 3.3 Alternative assessment specification

Checkpoints from Regular et al. (2025) Table 17: 2005 SSB = 26 kt (95% interval 22–31), 2017 = 451 kt (381–534), 2021 = 423 kt (NCAM and xteNCAM said to agree), 2024 = 342 kt (246–475), 2024/LRP = 1.24. The 2005 values (26 kt versus NCAM 25.18 kt) can agree on a low year without the two series sharing a reference point; agreement on a low year does not warrant splicing the two columns. The 2015 value is 33.8% of the old LRP, and the series is above the new one after 2016.



**Figure 5:** The two specifications. Overlap 1983–2015 RMSE = 126 kt (2015: NCAM 299 kt, xteNCAM 273 kt). Different safe sets. Not pooled.

**Table 6.** Rolling-origin RMSE on Specification B (kt). The persistence and training-mean rows are the origin-matched baselines, computed on the ladder’s own origin sets ( $n = 59$  at  $h = 1$ ,  $n = 55$  at  $h = 5$ ); the mixed-origin baselines over the longer naive origin sets ( $n = 63$  and  $n = 59$ ) are 88 and 318 kt for persistence and 449 and 506 kt for the training mean.

| Model   | $h = 1$   | $h = 5$    |
|---------|-----------|------------|
| persist | <b>84</b> | <b>300</b> |
| M1      | 120       | 432        |
| M1b     | 152       | 446        |
| M2      | 166       | 1059       |
| M3      | 127       | 930        |
| M4      | 206       | 1031       |
| mean    | 458       | 522        |

Collapse window (train 1954–89, test 1990–95): M1 RMSE 817 kt; M2 1898 kt. Official landings worsen the crash forecast. No module is retained. Persistence remains the lowest-RMSE forecast. This is a second non-retention outcome under the same rule on a different specification, and it is weaker than a statistical null result. The longer series and the revised LRP do not justify retaining the additional modules.

Recovery-stall window (train 1995–2012, test 2013–2024;  $n = 12$ ): M1 254 kt, M1b 178 kt, M2 269 kt, M3 268 kt, M4 269 kt. The 2013–2024 test path rises from 167 kt to a 2022 peak of 462 kt before easing to 342 kt in 2024, all above the 112 kt training-end value, so a persistence forecast frozen at that value errs by 262 kt — more than M1 and M1b on this single origin. This fixed-origin comparison does not enter the retention rule, which is decided on rolling-origin primary RMSE (Table 6, where persistence is not beaten at either horizon). M1b’s window fit carries the identification fragility familiar from its other window fits: a small fitted Allee parameter ( $s = 5.3 \times 10^{-3}$ ) with a carrying capacity ( $K = 500$  kt) extrapolated far beyond the training range (maximum 117 kt).

On the rolling Brier score for  $\mathbf{1}\{\hat{S} < \text{LRP}\}$  on Specification B (the secondary score, LRP = 276 kt): persistence, on its origin-matched sets,  $4/59 = 0.07$  at  $h=1$  and  $16/55 = 0.29$  at  $h=5$  (the same counts over the longer naive origin sets give  $4/63 = 0.06$  and  $16/59 = 0.27$ ); M1 0.05 and 0.45; M1b 0.15 and 0.47; M2 0.07 and 0.36; M3 0.02 and 0.31; M4 0.07 and 0.38. M1 and M3

improve the one-year Brier score over persistence (0.05 and 0.02 versus 0.07); no model improves it at  $h=5$ , and no model improves the primary RMSE score at either horizon (Table 6), so the retention verdict is unchanged. For reference, the same secondary score on Specification A is 0.00 at both horizons for persistence. The persistence indicator forecasts  $\mathbf{1}\{S_t < 884.6\}$  from the origin state, and its score is zero because every origin state is already below the 884.6-kt LRP. In particular the first origin,  $S_{1990}$ , lies below the 1983–89 mean bound, so the collapse train starts on the downslope of its own reference period. The target years 1991–2015 are below the LRP as well, so indicator and outcome agree at every origin. (The bare fact that targets are below the LRP would not by itself make the persistence indicator correct; the origin states being below it does.) Persistence supplies no directional prediction, so its sign-hit rate is reported as NA rather than as a zero hit rate. The structural scores are M1 0.04/0.05, M1b 0.00/0.05, M2 0.08/0.14, M3 0.08/0.10, M4 0.08/0.14 — none below persistence, which scores zero there by construction, and reported only to keep the two specifications’ records separate and complete.

Regular et al. assign the 1992–94 disappearance primarily to  $M$  (peak  $\approx 2.5$ ), informed by tagging and a capelin/cod predictor, and note that some of that  $M$  could still be unreported  $F$ . The relation between that attribution and the scalar residual of this ladder is stated in Section 3.2.

### 3.4 Prey-informed productivity

Murphy et al. (2025) report a 1991 acoustic collapse (1985–90 median 3704 kt versus 1991–2022 median 174 kt). A two-regime  $r$  with break 1991 uses the high regime only for forecasts issued before 1991. Capelin enters as a candidate disturbance class, not as a fitted parameter.

**Table 7.** Rolling RMSE, two-regime  $r$ . Baselines are origin-matched to the module’s own origin sets (Specification A  $n = 25/21$ ; Specification B  $n = 59/55$ ); the mixed-origin Specification B values over the longer naive sets are 88 and 318 kt.

| Specification | Model   | $h = 1$   | $h = 5$    |
|---------------|---------|-----------|------------|
| A             | persist | <b>98</b> | <b>265</b> |
| A             | M_cap   | 154       | 334        |
| B             | persist | <b>84</b> | <b>300</b> |
| B             | M_cap   | 147       | 894        |

On post-break origins only (Specification B,  $h = 1$ ;  $n = 33$ ): M\_cap 107 versus persistence 75 on the identical origin set (the all-origins value of 88 is quoted in the Table 6 caption). Not retained. On Specification A post-break origins ( $n = 24$ ) the corresponding M\_cap value is 149 kt.

The year-by-year 3L spring acoustic biomass is tabulated in the Northwest Atlantic Fisheries Centre’s Zenodo record (2025), 10.5281/zenodo.17515115, with 2023 = 331.3 kt from Murphy et al. (2025). Because  $(I_{\text{known}}/I_{\text{ref}})^b > 0$  for any positive index value and any sign of  $b$ , the scaled surplus keeps the sign of  $g(S_t)$ : the module can shrink production but cannot make it negative, so food limitation severe enough to drive a net loss lies outside what this specification can represent. Surplus is scaled by  $(I_{\text{known}}/I_{\text{ref}})^b$ , where  $I_{\text{known}}(t)$  is the last observation at or before  $t$ , subject to the regime rule of Section 2. The index is carried forward from the last observation available at the origin, and the carry is reset at 1991 only when the last observation predates the break: an origin at or after 1991 with no post-1991 observation yet is not evaluated. Origins before 1991 draw on pre-break surveys and are evaluated, so on Specification B the module is scored from 1988 ( $n = 36$  at  $h = 1$ ,  $n = 32$  at  $h = 5$ , of which three origins precede the break); on Specification A the first eligible origin is 1991 ( $n = 24$  and  $n = 20$ ). Training transitions with no eligible index value are dropped, and pre-1991 index observations are admissible in training for pre-break origins on the same rule. Along a multi-step trajectory  $t$  is the forecast year being stepped to, not the origin, so a five-year path may use index values published after its issue date;

this is a conditional hindcast with respect to the index and is not an operational forecast.  $I_{\text{ref}}$  is the training-window median of the observed index, and  $b$  is a third free parameter fitted jointly with  $r$  and  $K$  by one-step least squares on the training window. The module additionally faces degrees-of-freedom starvation: three parameters ( $r, K, b$ ) are fitted on as few as eight training years at the shortest origins, and after dropping years with no carried index value the shortest fit uses only six one-step transitions (origin 1991 on Specification A, 1988 on Specification B; the estimator refuses to fit on fewer) which invites in-sample overfitting and penalizes the one-year out-of-sample comparison. Survey years are unobserved in part of the record, so the index-forecast origins number  $n = 24$  ( $h = 1$ ) and  $n = 20$  ( $h = 5$ ) on Specification A and  $n = 36$  and  $n = 32$  on Specification B, against the SSB-based rolling-origin counts ( $n = 25$  at  $h = 1$  and  $n = 21$  at  $h = 5$  on Specification A; Section 4).

**Table 8.** Rolling RMSE, observed acoustic index. Bold marks the origin-matched baseline, which is the valid comparison for the module; the mixed-origin rows over the longer naive origin sets are retained for reference and are not the comparator. At  $h = 5$  on Specification A the module (262 kt) reads below the mixed-origin baseline (265 kt); that apparent 3-kt advantage is an origin-mix artefact and is not marked, because the origin-matched baseline on the module’s own origins is 193 kt. The comparison that matters is the matched one.

| Specification | Model                    | $h = 1$   | $h = 5$    |
|---------------|--------------------------|-----------|------------|
| A             | persist (mixed origins)  | 98        | 265        |
| A             | persist (origin-matched) | <b>97</b> | <b>193</b> |
| A             | M_cap_index              | 150       | 262        |
| B             | persist (mixed origins)  | 88        | 318        |
| B             | persist (origin-matched) | <b>79</b> | <b>288</b> |
| B             | M_cap_index              | 132       | 492        |

On the origin-matched reading the five-year near-tie on Specification A dissolves — the baseline on the module’s own origins reads 193 kt against the module’s 262 kt — and one-year RMSE is worse under both readings (150 versus 98 kt on the SSB origins and 97 kt on the module’s); on Specification B the module’s 132/492 kt stand against origin-matched baselines of 79/288 kt. The module is not retained under either baseline reading.

The scope of that result is set by the transmission the module encodes. The index scales contemporaneous surplus, so what is tested is whether prey information available at the origin improves the next steps through a proportional change in aggregate production. Published maturity schedules for this stock place the age at 50% maturity between ages five and six (Cadigan, 2016), so the pathway from a prey-driven change in recruitment to a change in spawning-stock biomass is delayed by roughly half a decade, and a contemporaneous multiplier cannot carry it. The same timing objection applies to M4, whose one-year stale start is an information delay rather than a recruitment delay. Non-retention here is therefore evidence about the tested transmission and its timing, not evidence that prey availability is unimportant to this stock.

### 3.5 Uncertainty on the retention margins

The uncertainty analysis attaches descriptive Diebold–Mariano loss-differential statistics (Diebold and Mariano, 1995; unweighted HAC truncation at lag  $h - 1$ ) and moving-block bootstrap intervals (Künsch, 1989; block length  $\max(h, 3)$ , 20,000 replications, seeded) to the retention margins of the retention rule. It is computed from the archived per-origin forecast files: on Specification B the xteNCAM rolling file; on Specification A the layer is computed on the primary coarse-regime pass. Its per-origin rows were regenerated from the registered estimator and archived; the regeneration reproduces the archived annual-landings per-origin file and the archived coarse-regime summary exactly, so no score changes. The layer

is unchanged when recomputed on the annual-landings pass. The persistence baseline is not archived per-origin; it is recomputed on the identical origin sets from the registered series, and the recomputation reproduces the recorded origin-matched baselines (98 and 265 kt on Specification A; 84 and 300 kt on the matched Specification B origins) — the script fails loudly otherwise. The moving-block intervals quantify sampling variation in the archived origin-level loss sequence conditional on the fitted forecast paths; they are not predictive intervals and do not propagate parameter-estimation, assessment-revision, catch-reconstruction or covariate uncertainty. The layer attaches uncertainty to margins the point rule has already ranked; it changes no frozen verdict, score, or table value.

The two columns of Table 9 summarise the same comparison through different approximations. The DM statistic  $z$  tests the mean of the per-origin squared-loss differential  $d_i = L_{A,i} - L_{B,i}$  against zero, scaled by a HAC standard error. The interval and  $p$  come from the moving-block bootstrap of the difference of RMSEs,  $\sqrt{\overline{L_A}} - \sqrt{\overline{L_B}}$ . The two effect measures are monotonically related, since  $\sqrt{\overline{L_A}} - \sqrt{\overline{L_B}} = (\overline{L_A} - \overline{L_B})/(\sqrt{\overline{L_A}} + \sqrt{\overline{L_B}})$ , so they share a sign and a zero null; the square root rescales the gap but does not by itself make one summary more stable than the other. What differs is the approximation used to gauge sampling variability — a HAC-scaled asymptotic normal reference against a resampling distribution over blocks — and in samples this small and this dependent the two need not agree. Four of the twenty-eight rows have an interval excluding zero while  $|z| < 1.96$  (Specification A  $h = 1$  M4 against M3; Specification B  $h = 1$  M3 against persistence; Specification B  $h = 5$  M1b against persistence and M4 against M3; see Section SI-3). We therefore report the two summaries side by side and do not treat the bootstrap interval as a second calibrated test. The interval and the  $p$  are mutually consistent in every row. The quantity reported as  $p$  is the percentile-tail fraction  $2 \min\{\#(\Delta_b^* \leq 0), \#(\Delta_b^* \geq 0)\}/B$  over the  $B$  block-resampled differences, obtained by inverting the percentile interval rather than from a null-centred resampling scheme; it is an interval-based summary and not a calibrated test of a sharp null, and a reported value of zero means no resampled difference fell on the opposite side of zero; where the two columns disagree both are reported and neither carries a verdict.

Because the origins overlap at  $h = 5$ , the estimation window expands, the predictand is a smoothed reconstruction and the sample straddles a structural break, the automatic  $h - 1$  HAC truncation and the  $\max(h, 3)$  block length should not be taken as neutral choices. The M1-against-persistence margin was therefore recomputed over a grid of both (Section SI-3). The DM statistic crosses the conventional threshold within the range of defensible bandwidths, upward on Specification A and downward on Specification B, so nothing rests on it; the bootstrap is markedly more stable across block lengths 3 to 9. The qualitative reading — Specification A within noise, Specification B separating — is therefore a bootstrap result rather than a DM result, established for that comparison and not asserted for every module. Read “within noise” as “this analysis does not resolve the difference”, not as evidence that the models are equivalent.

**Table 9.** Uncertainty on the retention margins. Both specifications are scored on their primary pass: Specification A rows are the coarse catch regime, so the RMSE values reproduce Table 4, and Specification B rows are the xteNCAM rolling pass. (Diebold–Mariano with HAC lag  $h - 1$ ; Künsch moving-block bootstrap, block length  $\max(h, 3)$ , 20,000 replications, seeded). Gaps are module minus comparator, in kt; positive gaps mean the module is worse.

| Spec | $h$ | Module | Comparator | $n$ | RMSE<br>module | RMSE<br>comp. | Gap (kt) | DM $z$ | 95% CI<br>(kt)    | $p$   |
|------|-----|--------|------------|-----|----------------|---------------|----------|--------|-------------------|-------|
| A    | 1   | M1     | persist    | 25  | 120.5          | 98.0          | +22.5    | 1.13   | [−13.4,<br>+52.2] | 0.192 |
| A    | 1   | M1b    | persist    | 25  | 114.8          | 98.0          | +16.8    | 0.98   | [−19.4,<br>+59.6] | 0.384 |
| A    | 1   | M2     | persist    | 25  | 144.1          | 98.0          | +46.0    | 1.29   | [−12.3,<br>+83.0] | 0.452 |

| Spec | $h$ | Module | Comparator | $n$ | RMSE<br>module | RMSE<br>comp. | Gap (kt) | DM $z$ | 95% CI<br>(kt)       | $p$   |
|------|-----|--------|------------|-----|----------------|---------------|----------|--------|----------------------|-------|
| A    | 1   | M3     | persist    | 25  | 134.6          | 98.0          | +36.6    | 0.99   | [-16.8,<br>+64.2]    | 0.932 |
| A    | 1   | M4     | persist    | 25  | 195.6          | 98.0          | +97.5    | 1.53   | [-9.1,<br>+172.7]    | 0.278 |
| A    | 1   | M2     | M1         | 25  | 144.1          | 120.5         | +23.5    | 0.91   | [-60.7,<br>+66.9]    | 0.684 |
| A    | 1   | M4     | M3         | 25  | 195.6          | 134.6         | +60.9    | 1.21   | [+4.4,<br>+134.4]    | 0.000 |
| A    | 5   | M1     | persist    | 21  | 288.7          | 264.7         | +24.0    | 1.84   | [-43.1,<br>+56.4]    | 0.279 |
| A    | 5   | M1b    | persist    | 21  | 288.6          | 264.7         | +23.9    | 1.84   | [-43.1,<br>+56.4]    | 0.284 |
| A    | 5   | M2     | persist    | 21  | 398.2          | 264.7         | +133.5   | 1.10   | [-21.0,<br>+217.5]   | 0.421 |
| A    | 5   | M3     | persist    | 21  | 366.4          | 264.7         | +101.6   | 1.12   | [-0.8,<br>+151.4]    | 0.078 |
| A    | 5   | M4     | persist    | 21  | 488.2          | 264.7         | +223.5   | 1.14   | [-20.8,<br>+409.4]   | 0.405 |
| A    | 5   | M2     | M1         | 21  | 398.2          | 288.7         | +109.5   | 0.98   | [-77.1,<br>+232.8]   | 0.805 |
| A    | 5   | M4     | M3         | 21  | 488.2          | 366.4         | +121.9   | 1.15   | [-27.4,<br>+263.2]   | 0.707 |
| B    | 1   | M1     | persist    | 59  | 119.5          | 84.4          | +35.0    | 2.81   | [+2.7,<br>+70.8]     | 0.032 |
| B    | 1   | M1b    | persist    | 59  | 151.6          | 84.4          | +67.2    | 3.60   | [+18.3,<br>+117.7]   | 0.009 |
| B    | 1   | M2     | persist    | 59  | 166.0          | 84.4          | +81.6    | 3.22   | [+36.4,<br>+122.8]   | 0.000 |
| B    | 1   | M3     | persist    | 59  | 126.7          | 84.4          | +42.3    | 1.85   | [+1.0,<br>+92.5]     | 0.042 |
| B    | 1   | M4     | persist    | 59  | 205.7          | 84.4          | +121.3   | 3.20   | [+53.7,<br>+193.0]   | 0.000 |
| B    | 1   | M2     | M1         | 59  | 166.0          | 119.5         | +46.6    | 1.78   | [-20.5,<br>+104.5]   | 0.170 |
| B    | 1   | M4     | M3         | 59  | 205.7          | 126.7         | +79.0    | 3.07   | [+35.6,<br>+116.4]   | 0.000 |
| B    | 5   | M1     | persist    | 55  | 431.9          | 300.0         | +131.9   | 2.06   | [+33.2,<br>+218.6]   | 0.008 |
| B    | 5   | M1b    | persist    | 55  | 445.5          | 300.0         | +145.5   | 1.80   | [+18.2,<br>+250.9]   | 0.023 |
| B    | 5   | M2     | persist    | 55  | 1058.9         | 300.0         | +758.9   | 2.15   | [+363.6,<br>+1038.5] | 0.000 |
| B    | 5   | M3     | persist    | 55  | 930.1          | 300.0         | +630.2   | 2.39   | [+352.8,<br>+845.6]  | 0.000 |
| B    | 5   | M4     | persist    | 55  | 1030.7         | 300.0         | +730.8   | 2.35   | [+407.3,<br>+978.3]  | 0.000 |
| B    | 5   | M2     | M1         | 55  | 1058.9         | 431.9         | +627.0   | 1.97   | [+195.7,<br>+929.7]  | 0.004 |
| B    | 5   | M4     | M3         | 55  | 1030.7         | 930.1         | +100.6   | 1.88   | [+20.2,<br>+177.4]   | 0.007 |

Splitting the same origins by period rather than by horizon shows what the pooled score aggregates. On Specification B at  $h = 1$ , computed on the archived per-origin records with no refit, persistence reads 94.7 kt before 1991 ( $n = 26$ ) against 142.4 kt for M1; 167.0 kt across 1991–1995 ( $n = 5$ ) against 126.8 kt for M1 and 66.6 kt for M1b; 20.7 kt across 1996–2012

( $n = 17$ ) against 19.7 kt for M3; and 61.0 kt from 2013 ( $n = 11$ ) against 96.4 kt for M3. These rolling scores answer a different question from the fixed-window collapse test of Section 3.1, where models fitted on 1983–1990 are asked whether a pre-collapse fit anticipates the crash, and none does. Here each forecast is issued from an origin already inside the decline, with training extended to that origin, so the question is whether the next year is closer to a refitted model or to the last observed state; a one-step forecast issued in 1993 from an already-collapsed state is not evidence that a model predicted the collapse. On that second question persistence is the lowest-error forecast in the two quiescent bands but not in the collapse band, where the structural modules read below it on five origins; leave-one-out on that band leaves M1 below persistence in all five cases, so the ordering is stable, though five origins remain too few to score. The 1996–2012 margin is 0.93 kt on a 20.67-kt baseline, inside the 5% tie band of 1.03 kt and with a bootstrap interval of  $[-14.2, +14.9]$  kt that includes zero: a tie under the stated rule, not a win. No retention verdict changes, because the rule scores pooled rolling RMSE at both horizons and the one sub-band advantage falls inside the band the rule declares non-deciding. The reading that changes is interpretive: the pooled result combines regimes in which persistence wins by a wide margin with a collapse band in which it does not, so the non-retention statement is a statement about the pooled score rather than about every period of this series.

Readings. On Specification A no non-retention margin against persistence separates from zero. At  $h = 1$  the deficits against persistence (16.8–97.5 kt on  $n = 25$ ) carry DM  $z$  from 0.98 to 1.53 with bootstrap  $p$  from 0.19 to 0.93; at  $h = 5$  (23.9–223.5 kt on  $n = 21$ )  $z$  runs from 1.10 to 1.84 with  $p$  from 0.08 to 0.42. The heavy-tailed collapse-window losses dominate the bootstrap variance at this sample size, so the Specification A outcome is a point-rule ranking whose margins lie within noise: the scores suffice to order the models and do not resolve small skill differences. On Specification B the non-retention separates: every module’s deficit against persistence has a bootstrap interval excluding zero at  $h = 1$  ( $p$  from 0.042 to below 0.001 on  $n = 59$ ) and at  $h = 5$  ( $p$  at most 0.023 on  $n = 55$ ). The declared (H1) comparators behave asymmetrically: M2’s deficit against M1 is within noise at  $h = 1$  on both specifications and at  $h = 5$  on Specification A, with the interval excluding zero only on Specification B at  $h = 5$  ( $p = 0.004$ ), while M4’s delay cost against M3 separates at  $h = 1$  on both specifications ( $p$  below 0.001) and at  $h = 5$  on Specification B ( $p = 0.007$ ) but not on Specification A ( $p = 0.71$ ). The alternative comparator rows are held in the archive rather than printed in Table 9. On the primary passes they read: Specification A  $h = 1$ , gap +29.3 kt,  $z = 0.91$ , interval  $[-67.6, +82.8]$  kt;  $h = 5$ , +109.7 kt,  $z = 0.98$ ,  $[-77.1, +232.8]$  kt; Specification B  $h = 1$ , +14.4 kt,  $z = 0.51$ ,  $[-67.5, +89.1]$  kt;  $h = 5$ , +613.4 kt,  $z = 1.93$ ,  $[+170.9, +945.1]$  kt. Accordingly, on them the reading (M2 against M1b) is within noise at  $h = 1$  on both specifications and at  $h = 5$  on Specification A, separating on Specification B at  $h = 5$  ( $p = 0.005$ ): the comparator declaration of the retention rule is immaterial to every recorded verdict, as the protocol disclosure records. The layer is produced by a registered script, deterministic under seed 0, with its output archived alongside it; Section SI-3 names both.

### 3.6 Fitted parameters

This section collects in one place every fitted-parameter value the article prints, the window or treatment it belongs to, and the bound it attains. The table collects values reported in the preceding sections; nothing is recomputed. Quantities not reported individually (the per-origin rolling fits of every module, the index module’s per-window exponent  $b$ , and the delay module’s structural setting) are held in the archive rather than invented. The collection exists so the identification record — bounds attained, interior fits, the flat valley — can be read at a glance.

**Table 10.** Fitted parameters, with the section in which each is reported.

| Module      | Window / treatment                                       | Fitted values                                                                                                                                                                                   | Bound / identification status                                                                                                        |
|-------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| M1          | Collapse, Specification A (train 1983–1990) (§3.1, §4)   | $r = 1.935$ , $K = 1032.7$ kt, constant catch 240 kt; repeller 144 kt, attractor 889 kt, monotone below 783 kt, $F'(S^*) \approx -0.39$                                                         | §3.1 records the collapse-window fitted $r$ as constrained near, but not attaining, the upper bound of 2                             |
| M1          | Recovery, coarse catch regime (§3.2)                     | $r = 0.458$ , constant catch 5.0 kt (the training mean); training MSE 128.35 kt <sup>2</sup>                                                                                                    | $K$ at 500.0 kt, the multi-start initialiser, not a bound (the lower bound on this window is $\max_{\text{train}} S + 10 = 50.8$ kt) |
| M1          | Recovery, annual landings (§3.2)                         | $r = 0.370$ , constant catch 3.19 kt (the 1995–2007 landings mean); training MSE 127.84 kt <sup>2</sup>                                                                                         | $K$ at its upper bound, 5000.0 kt                                                                                                    |
| M1b         | Recovery, coarse catch regime (§2.2, §3.2)               | $s = 2.1 \times 10^{-23}$ , $r = 2.000$ , $K = 105.8$ kt                                                                                                                                        | $s$ approaches lower bound to numerical tolerance; $r$ pinned at its upper bound in both treatments; $K$ a valid interior fit        |
| M1b         | Recovery, annual landings (§3.2)                         | $s = 9.4 \times 10^{-6}$ , $r = 2.000$ , $K = 129.8$ kt                                                                                                                                         | $s$ at the lower boundary to numerical tolerance                                                                                     |
| M1b         | Recovery-stall, Specification B (train 1995–2012) (§3.3) | $s = 5.3 \times 10^{-3}$ , $K = 500$ kt, training-window maximum 117 kt                                                                                                                         | identification fragility: $s$ small, $K$ extrapolated far beyond the training range                                                  |
| M3          | Rolling, Specification A (§3.1)                          | $\phi = 0.95$ (AR(1) residual coefficient)                                                                                                                                                      | printed value; the per-window rolling $\phi$ values are archive items                                                                |
| M_cap_index | Index module (§3.4)                                      | $r$ , $K$ , $b$ fitted jointly by one-step least squares                                                                                                                                        | fitted values not printed (archive); degrees-of-freedom note at $n = 8$ training years                                               |
| M1          | Flat-valley sweep, recovery window (§3.2)                | $K$ fixed anywhere on [60, 5000] kt moves the one-step training objective only from MSE 127.4 to 149.9 kt <sup>2</sup> (training RMSE 11.29–12.24 kt) while $r$ compensates over [0.435, 0.773] | the $(r, K)$ pair is not identified on this window; the ordering of valley variants is not a robust ranking                          |
| M4          | Rolling, all specifications                              | $r$ , $K$ , $\phi$ refitted by the same procedure as M3 at every origin; the module adds no parameter of its own, the one-year delay being structural                                           | archive: the refitted per-origin values coincide with M3's and are not printed separately                                            |

| Module      | Window / treatment      | Fitted values                                                                                                                                   | Bound / identification status |
|-------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| All modules | Per-origin rolling fits | archive items (not printed); M1b's Specification B rolling Allee optimum is environment-sensitive at the $\pm 17$ kt level (§Data availability) | —                             |

Bounds:  $r \in (0.001, 2]$ , and  $K$  optimised on  $[\max_{\text{train}} S + 10, 5000]$  kt, where the maximum is taken over the states the one-step regression conditions on. On the 1995–2007 recovery window those are the years 1995–2006, with maximum 40.8 kt, giving a lower bound of 50.8 kt. Optimisation starts from  $K = 500$  kt. The recovery-window M1 fit reports  $K = 500.0$  kt: the optimiser does not move from its starting value, which lies 449 kt above the bound, because the objective is flat there. This is the same flat-valley geometry quantified in Section 3.2, and a stationary starting value is a sharper symptom of it than an active bound would be.

### 3.7 Operating characteristics of the retention rule

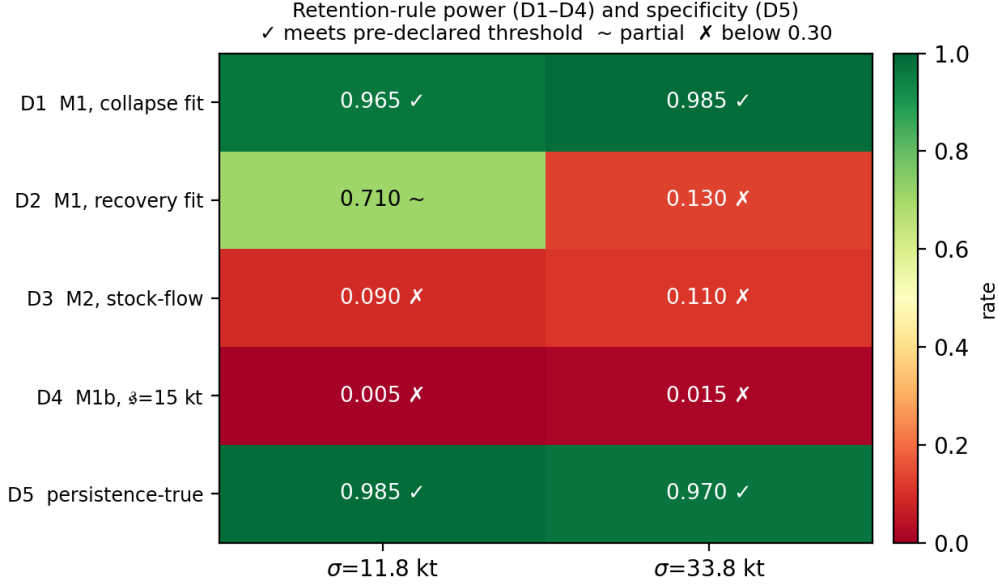
The result above is negative, so its interpretation turns on whether the rule could have detected added structure had it been present. That question cannot be answered from the observed series alone. It is addressed by simulation, under a design — data-generating processes, replicate count, and the thresholds separating an adequate from an inadequate rule — fixed in advance of generating any synthetic series.

Five processes were simulated, each a member of the ladder's own class and parameterised from the fitted values of Section 3.6, using the estimation and trajectory code of Section 2.3 unmodified: an autonomous Schaefer map at the collapse-window fit (D1) and at the recovery-window fit (D2), a stock-flow map with the coarse regime path (D3), a depensatory map with an identifiable threshold  $\mathfrak{s} = 15$  kt (D4), and a persistence-true null with no surplus term (D5). Each was run at  $\sigma = 11.8$  and 33.8 kt, the archived recovery- and collapse-window residual standard deviations, on 33-year series matching Specification A, with 200 seeded replicates per cell. The full retention rule was applied unchanged.

| Process                                | Generating module | $\sigma = 11.8$ | $\sigma = 33.8$ |
|----------------------------------------|-------------------|-----------------|-----------------|
| D1 autonomous, collapse fit            | M1                | 0.965           | 0.985           |
| D2 autonomous, recovery fit            | M1                | 0.710           | 0.130           |
| D3 stock-flow                          | M2                | 0.090           | 0.110           |
| D4 depensation, $\mathfrak{s} = 15$ kt | M1b               | 0.005           | 0.015           |
| D5 persistence-true                    | none retained     | 0.985           | 0.970           |

The first four rows give the proportion of replicates in which the generating module was retained; the last gives the proportion in which no structural module was retained when persistence was true. The false-retention rate across all structural processes was 0.044.

Two properties follow. The rule is *specific against in-class alternatives*: under a persistence-true process drawn from the ladder's own family it declines to retain structure in 97–99% of replicates, so the empirical non-retention is not an artefact of an instrument that rejects everything. That specificity does not extend beyond the class. Two further processes were simulated whose truth the ladder cannot represent — productivity declining smoothly with time, and a deterministic trajectory observed with error — and the rule retained a structural module in 68% to 98% of replicates, almost always the autonomous map. Against a truth outside its class the rule over-retains, and the specificity figure above should be read as conditional on



**Figure 6:** Operating characteristics of the retention rule. Cells give the proportion of replicates in which the generating module was retained (D1–D4) or in which no structural module was retained under a persistence-true process (D5), against the thresholds fixed before execution: a tick marks power at or above 0.80 and specificity at or above 0.90, a cross marks power below 0.30. Power is adequate only where the generating signal is strong (D1); the rule is specific throughout.

the class rather than as a property of the rule. It is also *conditionally powerful*: where the generating signal is strong, as in the high-biomass, large-catch conditions of D1, it recovers the true module in 97–99% of replicates.

Power is nevertheless low for three of the four structural processes, and the binding constraint is identification rather than the rule’s gates. Requiring a module only to beat persistence, ignoring the comparator and the band, still retains the true module in just 33% of D3 and 18% of D4 replicates; conditional on clearing that bar the comparator requirement then removes a further 69% and 94%. With five modules, chance alone would place the generating module first 20% of the time, and on D3 it holds that position in only 3.8% of replicates — less often than a random draw, with a mean one-year error above that of two rival modules. Section SI-5 gives the full decomposition.

The consequence for the empirical result is that its strength is not uniform across modules. Had the collapse window been generated by an autonomous map at the fitted parameters, the rule would have retained M1 in 97% of replicates; it does not retain M1 on the observed series, so those data are inconsistent with that generating process and the non-retention is evidence about the stock rather than about the instrument. Non-retention of the stock-flow and depensation modules under recovery-window conditions is by contrast weak evidence against those structures, because the rule would have retained them in at most 11% and 1.5% of replicates even had they been true. The residual and lagged-initialisation modules were not simulated as generating processes, so no power estimate is available for them; their power is plausibly bounded by the stock-flow figure, since both extend the same base map and must clear an additional comparator, but that is an inference rather than a measurement.

Two limits are inherent to the design. The processes are members of the ladder’s own class, so this measures power against correctly specified alternatives and the D1 figure is an upper bound; and the synthetic predictand carries no assessment smoothing, so persistence is a weaker baseline here than on the reconstructed series.

The same ladder has been scored on a groundwater system, the Edwards Aquifer, in Abaee

(2026, Does a one-pool water-balance model improve forecasts of Edwards Aquifer head), and the comparison isolates a property that this series cannot show. That analysis applied the retention rule without a tie band and horizon by horizon; the figures below restate its verdict under the rule used here, so that both systems are judged by one criterion. The restatement changes no outcome: the retained set is empty on both systems under either form of the rule. Six Edwards margins fall inside the 5% band, and the module they bear on was already excluded on other grounds. There the autonomous module reads below persistence on the point ranking at one year (12.84 against 13.23 ft) and is still not retained, because the margin of 2.96% falls inside the band and the ordering reverses at five years (21.25 against 21.11 ft). On the present series no module comes within the band at either horizon: the closest approach is M1b at 9.0% adrift at five years and 17.1% at one year, against a band of 5%. The two cases therefore reach the same verdict by different routes: here the ranking alone decides, whereas on the groundwater series the tie band and the two-horizon requirement are what withhold retention from a module the point rule would have kept. Neither series is pooled with the other and no verdict is transferred; what recurs is the decision rule, not the data or the outcome.

Because the simulation archives every replicate’s scores, alternative decision rules can be evaluated on the same synthetic data without refitting. Averaging over the four structural processes, and over the persistence-true process for specificity:

| Decision rule                                   | mean power | specificity |
|-------------------------------------------------|------------|-------------|
| Retention rule as adopted (H1–H3, 5% band)      | 0.376      | 0.978       |
| Without the comparator gate (H2, H3 only)       | 0.476      | 0.972       |
| Persistence only, $h = 1$ , 5% band             | 0.562      | 0.955       |
| Persistence only, any margin                    | 0.542      | 0.765       |
| Full rule, no tie band                          | 0.436      | 0.772       |
| Full rule, 10% band                             | 0.337      | 0.998       |
| MASE < 1 against the naive benchmark            | 0.651      | 0.675       |
| Information criterion, $n \log \text{MSE} + 2k$ | 0.509      | 0.992       |

The last two rows place the rule against selection criteria in common use. Scoring by scaled error alone raises power to 0.651 but drops specificity to 0.675; an information criterion penalising free parameters does better than the rule adopted here on both axes, reaching 0.509 power at 0.992 specificity. A criterion penalising parameter count therefore dominates the rule applied here on both axes, and the tie band is a coarser device for the same purpose. Within the rule as specified, the band carries most of the weight — removing it costs 0.206 of specificity to buy 0.060 of power — while the comparator gate costs 0.100 of power for 0.006 of specificity. A diagnostic that identified in advance where the rule has power would be more useful than any variant, and Section SI-5 reports two candidates that fail: the better-correlated one is computed from the same scores the rule consumes and misclassifies both stock-flow cells. Identifying when the rule has power remains open.

A module may improve the score and still not constitute added structure. In the companion, the groundwater stock-flow variant fitted with training-mean fluxes beats persistence at one year yet was declined on the ground that it collapses to an autoregression when those fluxes are held constant. M1b raises the same question in a different form. Its fitted threshold descends towards an unattained infimum, at which point the production term is the zero-threshold cubic branch of Definition 2.1 rather than a positive-threshold Allee model, so its lower error on Specification A is not evidence for depensation. The scored rule does not encode this distinction — it compares errors — and both cases are resolved by inspecting what the fitted module has become, not by the score.

### 3.8 The outcome years (2016–2021) on the Specification-A horizon

Specification A is scored on 1983–2015. The assessment of record for the same model family was subsequently updated, and the per-year continuation of the series through 2021 is published in the RAM Legacy Stock Assessment Database (v4.66, stock COD2J3KL; the 2021-vintage reading of the same NCAM-family assessment). Table 16 reports the outcome years under both published vintages, with the persistence error from the 2015 origin.

| Year | RAM v4.66 SSB (kt) | xteNCAM SSB (kt) | Persistence error (kt) |
|------|--------------------|------------------|------------------------|
| 2016 | 340                | 339              | 63                     |
| 2017 | 433                | 451              | 156                    |
| 2018 | 394                | 369              | 117                    |
| 2019 | 419                | 400              | 142                    |
| 2020 | 440                | 414              | 163                    |
| 2021 | 411                | 423              | 134                    |

**Table 16:** The outcome years on the Specification-A horizon. Persistence error is  $|S_t - S_{2015}|$  from the RAM-vintage origin of 277 kt; the xteNCAM column is the independent model’s reading of the same calendar years.

Three facts are relevant to the frozen test. First, the outcome window is an increase regime: every outcome year lies above its 2015 origin in both vintages, so the window contains no collapse event of the kind the fixed pre-collapse origin test was designed to probe, and the frozen verdict is neither confirmed nor reversed by it. Second, the two vintages read the same years within 1–26 kt (mean absolute difference 16.8 kt), so the outcome is vintage-robust even where the historical part of the series is not (Section 3.9). Third, persistence errs low throughout: from the 2015 origin the rule that nothing changes under-predicts every outcome year, with mean absolute error 129.2 kt and RMSE 133.3 kt on the RAM vintage (107.5 kt mean absolute error from the Table A2 origin of 298.65 kt) — the same order as the scored one-step persistence RMSE of 98 kt on 1983–2015. Re-scoring the module ladder on the extension is archived pipeline work under the frozen specification and runners of the reproducibility statement; this section claims the outcome facts and the baseline errors, and nothing about module rankings on the extension.

### 3.9 Threshold definition-dependence on the published series

The two specifications use different thresholds on different series. The published files allow the stricter experiment: both thresholds on one series. On the xteNCAM file (1954–2024; its ratio column fixes  $B_{\text{lim}}$  at 276 kt, with per-year implied values 250–289 and mean 275.5), classifying each year under  $B_{\text{lim}} = 276$  kt and under the 2010/2016 LRP of 884.6 kt gives 45 years at or above  $B_{\text{lim}}$ , 16 years at or above the LRP (1954–1969 only), and 29 disagreement years in two blocks: 1970–1992 without 1977–1979, and every outcome year 2016–2024. Three readings follow. (i) Under the 1980s-mean threshold the stock stands below the reference point for 55 consecutive years (1970–2024); under the  $B_{\text{lim}}$  threshold it is below for 26 years (1977–1979 and 1993–2015) and above from 2016 onward. The late-1970s episode (ratios 0.69–0.95) is a critical-zone reading, on the current model, of years the contemporary fishery regarded as healthy. (ii) The threshold itself is vintage-dependent: the 1983–1989 mean that defines the LRP equals 884.6 kt on the Table A2 vintage and 790.0 kt on the RAM vintage — the reference point moves by 94.6 kt across vintages of the same assessment family. (iii) The verdict on identical data is therefore threshold-relative in both senses, of definition and of vintage; the October 2023 framework revision, which moved the stock’s reported zone from critical to cautious without a change in the stock, is this experiment executed by the assessment authority itself. The scored RMSE comparison is untouched — it never uses the thresholds — but every normative verdict

layered on the forecasts, the Brier indicator score included, inherits the dependence.

### 3.10 The 2022–2023 observation suspension as a review-timing event

The frozen test conditions on a complete annual series. The stock’s real 2022–2023 history violates that condition. Partial coverage by the fall research-vessel survey in 2021 and 2022 left the leading indicator of stock size uncomputable; no typical assessment update was possible in 2022 or 2023; the stewardship maximum was rolled over unchanged at 12,999 t; and the survey vessel that carries the series was condemned in February 2023. Table 17 assembles the episode from the assessment of record and the public record.

| Period   | Event                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------|
| 2021–22  | Partial fall-survey coverage; leading indicator not computable                                            |
| 2022–23  | No typical assessment update; stewardship maximum held at 12,999 t                                        |
| Feb 2023 | Survey vessel condemned                                                                                   |
| Mar 2024 | Revised-framework assessment: SSB at $1.2 B_{\text{lim}}$ (95% CI 0.7–2.1; 22% critical-zone probability) |
| Jun 2024 | Commercial fishery reopened at 18,000 t after 32 years                                                    |
| 2025     | TAC 38,000 t; reference point revised after being deemed overestimated                                    |
| Apr 2026 | Healthy-zone report: SSB $\approx$ 540 kt (CI 420–700); 70% probability above the upper stock reference   |

**Table 17:** The 2022–2023 observation suspension and its resolution (sources: Regular et al. 2025, Table 17 file; DFO reopening release 2024; the 2025 and 2026 assessment coverage).

Read against the review-timing rule — a review must land no later than the worst-case loss of the information it consumes — the episode is that rule’s boundary case in the wild: the information loss ran past the review cadence, and the removals rule froze for two cycles while the observation map was dark. The companion monitoring paper states the formal identity (viability at the deadline requires the review no later than worst-case exit, boundary inclusive); the 2022–2023 suspension is its empirical instance, and it marks the practical limit of any scoring exercise that presumes the series exists.

## 4. Discussion

Section 3.7 separates two readings of the empirical result. Where the rule has power, non-retention is evidence about this series; where it does not, non-retention constrains little. The discussion below is scoped accordingly: mechanisms are proposed for the autonomous module, whose non-retention the simulation shows to be informative, and withheld for the stock-flow and depensation alternatives, whose are not.

The observed path falls far below the fitted lower equilibrium and then recovers. No exact fixed autonomous noise-free scalar trajectory can reproduce that. The obstruction is structural, and is stated below for the map held at constant catch; the scored collapse failure has a simpler and more direct cause, given after the proposition.

**Proposition 4.1 (No recovery from below the lower equilibrium, for autonomous noise-free scalar surplus maps).** *Let  $S_{t+1} = F(S_t)$  be an autonomous one-dimensional surplus-production map of Definition 2.1 evaluated with  $\varepsilon_t \equiv 0$  and a constant catch  $C_t \equiv C$ , with at most two positive equilibria — a lower repelling point  $S_-$  and an upper attracting point — and a one-step map monotone below its maximum; the single-equilibrium monotone regime is the special case, and the fitted collapse-window map ( $r = 1.935$ ,  $K = 1032.7$  kt, constant catch 240 kt) has exactly this form (repeller 144 kt, attractor 889 kt, monotone below 783 kt). If the observed path  $(S_t)$  falls strictly below  $S_-$  at some time and is strictly larger at some later time, then no trajectory of  $F$  reproduces  $(S_t)$  exactly. The Specification A collapse path satisfies the*

*hypothesis: NCAM SSB falls to 9.7 kt by 1995, far below the fitted repeller of 144 kt, and rises thereafter.*

The argument is immediate: below the lower equilibrium  $F(S) < S$ , and  $[0, S_-)$  is forward-invariant because  $F$  is monotone there and  $F(S_-) = S_-$ , the interval being closed at the left because the implementation clips the state at a non-negative floor. Every trajectory entering  $[0, S_-)$  therefore decreases toward the floor and never returns above  $S_-$ , so a path that goes below  $S_-$  and is later higher is not a trajectory of  $F$ .

Two limits on the scope of this proposition should be stated explicitly, because both are easy to overstate. First, it does not say that autonomous maps cannot produce non-monotone paths: the fitted map has  $F'(S^*) \approx -0.39$  at its attractor, so approach to the attractor is by damped oscillation, and declines followed by recoveries are the generic local behaviour there (from 950 kt the fitted map gives  $950 \rightarrow 857 \rightarrow 899$  kt). What is excluded is recovery after a fall below the lower repeller, which is the feature the observed collapse path actually has. Second, the result is a statement about autonomous noise-free scalar maps only. It does not apply to the stochastic map of Definition 2.1 with  $\varepsilon_t \neq 0$ , nor to M3 and M4, which carry a residual state and a stale initial state respectively and are therefore not autonomous scalar maps. The proposition applies directly to M1 under the deterministic forecast convention used here, and to M1b whenever its own fitted map has the stated equilibrium structure. It does not formally cover M2, whose prescribed year-by-year catch makes the update a time-indexed family  $F_t$  rather than a single autonomous map; for M2 the argument is motivating rather than binding, and the collapse-window result of Section 3.1 is the relevant evidence. It is not an impossibility theorem for surplus-production modelling in general.

A third limit is the most consequential for ecological reading, and it follows from the same algebra. The lower equilibrium is generated by the removals, not by any compensatory term in the production function. For the Schaefer map the positive equilibria of  $rS(1 - S/K) = C$  are  $S_{\pm}(C) = (K/2) \left[ 1 \pm \sqrt{1 - 4C/(rK)} \right]$ , so the repeller is a function of catch: at the fitted  $r = 1.935$ ,  $K = 1032.7$  kt it sits at 144 kt under  $C = 240$  kt, at 66 kt under  $C = 120$  kt, and at 2.6 kt under the moratorium-level  $C = 5$  kt, approaching zero as  $C$  does. The 1995 SSB of 9.7 kt is far below the first of these and above the last. The obstruction is therefore a statement about a stock held under sustained removals of the pre-collapse magnitude; it is not evidence of biological depensation, and it does not explain non-recovery after catches fell. Nothing in Definition 2.1's Schaefer branch contains an Allee effect — that is the separate M1b branch — so a threshold recovered from an M1 fit should not be read as a biological one.

The collapse forecasts fail for a more direct reason than the equilibrium structure. Every module carries stationary production, so each predicts positive surplus at the biomass levels of the late 1980s while the reconstruction falls. For the modules supplied with the catch path — M2, M3 and M4 — the 1992 decline in prescribed catch raises the predicted increment directly, so they turn upward exactly where the reconstruction turns down; M1 and M1b use a training-mean constant and turn upward because that constant is smaller than the losses the reconstruction records. That mechanism, not the crossing of a catch-dependent repeller computed at  $C = 240$  kt, is what the fixed-origin collapse scores record.

The fixed-window sign-hit rates of  $\Delta S$  for the structural modules (0.00–0.50 on collapse; NA for the baselines, which supply no directional prediction; Table 3) record the directional character of that failure: the structural models miss not only the magnitude of the crash but the sign of the trajectory exactly when the official catch series drops.

Under both the coarse catch regime and official landings, lowering  $C$  in 1992 cannot generate the crash. Neither prescribed catch treatment reproduced the reconstructed decline in these implementations; which process accounts for the shortfall — productivity, unallocated mortality, or the observation and reconstruction of the series — is not identified by this comparison. Nor does this test whether fishing caused the historical collapse: in this accounting  $S_t$  is spawning-stock biomass while  $C_t$  is total landings, which is not an age-structured removal from the

spawning stock, so the exercise bounds a scalar catch-driven surplus story and not the causal role of the fishery. That is compatible with DFO’s caution that NCAM  $M$  can absorb unreported deaths: the crash remains such an event under the best available public catch reconstruction, which tightens that limitation rather than relaxing it.

Unidentified extra structure does not close the gap to persistence, and in two of the five structural modules (M2 and M4) it widens it. The direction is not uniform and should not be stated as though it were: relative to its own declared comparator M3 reads below M2 in all six rolling comparisons (by 6 to 129 kt), and M1b reads below M1 at  $h = 1$  on Specification A and on the fixed recovery window, though not on Specification B, where M1b is the higher of the two (152 against 120 kt at  $h = 1$ ). What fails is H2, not always H1. Where the added structure does hurt, the mechanism is visible: autoregressive residuals fitted on short, regime-changing windows persist the wrong sign, an Allee parameter goes to the boundary, and a one-year delay removes information. These modules stay descriptive unless the conditions under which a module’s contribution could be certified rather than merely scored hold. The delay mechanism itself is formalized exactly: the certification companion’s delayed-information theorem bounds guaranteed recognition time by the pre-observation state (Abae, 2026, An obstruction calculus), the theorem-level form of the one-year lead documented here.

The 2016 LRP is the 1980s mean SSB, so over its own reference years the mean deviation of SSB from that bound is zero by construction, though individual annual deviations are not. Defining the threshold from the same years that a leading-indicator claim would be tested on does not establish prospective early-warning performance for 1983–1990.

The moratorium is a change in implementable catch and is already in  $C_t$ . A separate delay switch does not add a degree of freedom beyond stock-flow. Whether the 1992 management architecture admits any feasible catch path that keeps the stock above its reference threshold is a separate question from one-year forecast error, treated for this stock in Abae (2026, Robust viability of the 2J3KL limit reference point), and is not addressed here.

One-dimensional surplus production is not NCAM: age structure, migration, and survey catchability are omitted. The test is whether this ladder is retained under the stated score, not whether the assessment is a good filter. M4 is a stale-start experiment rather than a fully delayed information set: its trajectory begins at  $S_{t-1}$ , but its parameters are M3’s, fitted on training transitions that end at  $S_t$ , so the later state still reaches the forecast through the fitted coefficients and the carried residual. Because the path starts one year earlier, reaching a target  $h$  years after the origin takes  $h + 1$  updates rather than  $h$ . The catch path is indexed from the earlier start, so the first update applies  $C_{t-1}$ , and the residual recursion is applied before the first step, so the first projected residual is  $\phi e_{\text{last}}$  where  $e_{\text{last}}$  is the residual of the transition  $S_{t-1} \rightarrow S_t$ ; the first update therefore re-forecasts the transition on which that residual was fitted. It is therefore a perturbed initialisation evaluated at calendar time  $t$ , not propagation from a genuinely year-old information set. A genuinely delayed variant would truncate training as well, and is not scored here. Its one-year gap against persistence therefore mixes a stale initial state with the model’s own cost.

The separating control — persistence issued from the last available assessment,  $S_{t-1}$  — decomposes the gap. On Specification A under the coarse catch regime (the primary pass) the control reads 184.4 kt at  $h = 1$  and 329.8 kt at  $h = 5$ , against M4’s 195.6 kt and 488.3 kt. The information delay, the year-old start under the same persistence rule, therefore accounts for 86.4 of the 97.5-kt one-year gap and the surplus model’s own cost for the remaining 11.1 kt; at  $h = 5$  the split is 65.1 against 158.4 kt.

On Specification B every term is computed on M4’s own origin set, so that the control, the baseline and the module share origins ( $n = 59$  at  $h = 1$ ,  $n = 55$  at  $h = 5$ ). Matched persistence reads 84.4 and 300.0 kt and the stale-start control 158.0 and 337.4 kt, against M4’s 205.7 and 1030.7 kt. The delay accounts for 73.6 of the 121.3-kt one-year gap and the model’s own cost for the remaining 47.7 kt; at  $h = 5$  the model’s own cost dominates (693.3 of 730.8 kt). These are

descriptive comparisons between two forecast rules, not a causal decomposition of the value of assessment timeliness: the delayed control still reads off the same smoothed reconstruction, and access to a later reconstruction is not the same thing as access to a timely real-world assessment. With that reading, M4’s error is numerically dominated by the initialisation component at the one-year horizon on both specifications, and mostly the surplus model’s own structure at the five-year horizon on Specification B: at  $h = 1$  the surplus model’s own penalty is only about 11 kt — relative to the stale-start control the gap decomposes numerically into an initialisation component and a residual-surplus component, an arithmetic split between forecast rules rather than a causal attribution. For reference, the one-year lag in the start state accounts for 86.4 kt at  $h = 1$  on Specification A, against structural costs over timely persistence of 23 kt for M1, 17 for M1b, 46 for M2 and 37 for M3, and M4’s own structure-given-delay cost of 11.1 kt. These are differences between forecast rules on a smoothed reconstruction and are not an estimate of the operational value of assessment timeliness.

The predictand is an assessment smoother (NCAM/xteNCAM SSB), not a raw observation: forecasting it is forecasting a filtered state, and persistence inherits the smoother’s autocorrelation — the comparison is valid for “does this ladder track this assessment,” not for a vintage or early-warning reading, which would require assessment vintages at each origin, and none is available. Persistence is correspondingly not an empty null on this predictand. Spawning-stock biomass aggregates surviving mature cohorts of a long-lived species, and the assessment adds further temporal smoothing, so strong one-year autocorrelation is the expected baseline on biological and observational grounds alike. Beating it requires useful prediction of subsequent change, whether a sustained trend or a transition such as the collapse, the onset of rebuilding, or the post-2015 stall. Persistence is retained as the scoring benchmark the retention rule is defined against, not as a biological projection, and no demographic balance is asserted on its behalf at either horizon. Recreational catch remains incompletely measured.

The log-RMSE scores of Table 4 carry a floor caveat: structural trajectories absorb at the numerical floor  $\varepsilon_{\log} = 10^{-3}$  kt rather than at zero (the trajectory code clips the state to  $[\varepsilon_{\log}, 10^6]$  kt), and the reported values use  $\log \max(\hat{S}, \varepsilon_{\log})$ ;  $\varepsilon_{\log}$  is part of the registered scoring configuration and is distinct from the process noise  $\varepsilon_t$  of Definition 2.1. The floor binds often — for M1 and M1b it binds on a majority of five-year origins on both specifications — and the per-origin counts are given in Section SI-4. The raw-RMSE column is the retention score, so no retention verdict depends on the floor; it affects the log column only, which is reported as a secondary diagnostic. Recomputing that column with the floor lowered to  $10^{-6}$  and  $10^{-9}$  leaves the log-RMSE ordering of the five modules unchanged at both horizons on both specifications, so no secondary ranking depends on it either. A constant-productivity map is misspecified for a series whose assessment attributes the crash to  $M \approx 2.5$ ; the test is whether that misspecification still forecasts SSB better than persistence, and the scored comparison answers that for this ladder on these series. Sample sizes are small ( $n = 33$  years on Specification A; rolling  $n = 25$  at  $h = 1$  and  $n = 21$  at  $h = 5$ ). They suffice to rank models and do not suffice to certify a small skill difference. The 2023 40%  $B_{\text{MSY}}$  LRP is the Specification B setting, and Section 3.3 is that review: the ladder was run on the xteNCAM series under a fresh specification-matching review, with the same non-retention verdict.

**Reconstruction-level corroboration.** An independent, assessment-level record sharpens the same boundary. Rose (2026) compares two surplus-production reconstructions of the stock spanning 1983–2023 — the Rose and Walters (2019) model and the DFO (2024a) assessment model — and documents that surplus production and stock growth stalled after 2015, with some years negative, the stall-point biomass being controversial between the two reconstructions and well below historical norms. The recovery-stall window of Specification B (train 1995–2012, test 2013–2024) overlaps that period (its 2016–2024 portion). A stock whose surplus production and stock growth stall together, with some years negative and the stall-point biomass below historical norms — the configuration Rose (2026) reports — is precisely the configuration a

constant-productivity surplus law cannot reproduce, and the ladder’s single-origin comparison on that window is consistent with it on the rolling score, which is what decides retention. On that single fixed origin M1 (254 kt) and especially M1b (178 kt) do read below frozen persistence (262 kt); the retention verdict is nonetheless negative, because it is decided on rolling-origin RMSE, where persistence is not beaten at either horizon. The fixed-window result is reported because selecting favourable episodes would tell a different story from rolling evaluation, and that contrast is itself part of the finding. The two reconstructions’ disagreement at the stall point is moreover the assessment-level twin of the ladder’s own non-uniqueness: M1b’s apparent recovery-window skill is carried by a carrying capacity extrapolated far beyond the training range, exactly the extrapolation on which the two reconstructions diverge. Rose (2026) also documents the limit-reference-point trajectory — successive downward revisions to just over 0.3 Mt, then to about 0.25 Mt in 2025 — which is the norm-field drift that separates the two specifications evaluated here (884.6 kt versus 276 kt) and the reason their columns are never pooled.

**Attribution and the prey modules.** Rose (2026) infers from structural-equation models that the post-2015 production deficit is limited by capelin and amplified by harp seal predation, with fishing almost certainly not the sole cause. That attribution is consistent with the ladder’s two negative findings, and it sharpens both. First, it matches the catch-treatment null: neither prescribed catch treatment repairs the collapse, which is what one would expect if the dominant missing term is not catch. Both treatments could omit relevant removals, and other structural errors could prevent either from helping, so the scored result is consistent with that attribution rather than establishing it. Second, it draws the identification boundary inside the prey results: capelin limitation can be a real production driver — the 1991 acoustic collapse is large (1985–90 median 3704 kt versus 1991–2022 median 174 kt) — while the corresponding forecast modules still fail the stated score. A driver’s reality at the reconstruction level does not make it a usable forecast module at the surplus-production level: the two-regime and acoustic-index variants are weakly constrained by their short training windows and missing index years, and they fail the retention score. No parameter profile was computed for them, so their non-retention rests on prediction error rather than on a demonstrated identification failure.

**Protocol status.** The evaluation windows and the scoring core — the primary RMSE comparison and the persistence bar — were coded before the first scoring pass and applied unchanged. Two components of the retention rule were not: the 5% tie band and the comparator declarations for the non-nested rungs were formalised after the scores of Section 3 had been computed. Neither affects any outcome reported here; the tie band never decides a retention result, the smallest deficit being 17.1%. The design is therefore a fixed computational protocol, not a prospective registration, and the additional passes (annual landings, survey start, capelin, Specification B) were declared in the text as they were added. No dated pre-scoring protocol file exists for this study. Section SI-1 records the order in which each pass was run.

The results are specific to the objects tested. They do not extend to other estimators of the same model family, to closed-loop filtering schemes, or across the two assessment specifications, which are scored separately throughout. The delay evaluated here is a one-year information delay in the forecast start state, not a continuous-time retarded system.

On this evidence, within the scope of this estimator and ladder, modules that are not identified on the training data did not reduce forecast error below matched persistence at both horizons here. Weak identification is documented for the recovery-window Allee fits and is not established as the sole cause of non-retention. The retained forecast is persistence.

Sections 3.8–3.10 locate the practical margin. The outcome years contain no event the ladder could win, the verdict layer moved under the stock twice in a decade (threshold revision, vintage revision), and the one genuine decision failure of the period was an observation failure. On this stock, the binding constraint is the monitoring and reference-point layer, not forecast structure; the retention rule’s empty set is a statement about structure, and the monitoring companion’s

design rules are the instrument matched to where the loss of value actually occurred.

**A mechanized note on the ladder bound.** The ladder inequality used to bound the retained set is formalized in the project’s Lean 4 layer, in namespace `Formalizations.E1` (`E1_ForecastLadder`): `ladder_telescope`, `ladder_bound_upper`, `ladder_bound_lower`, `ladder_bracket` and `ladder_ascent`, stated over an ordered-field interface. A comment-stripped source scan of the project finds no `sorry`, `admit`, `axiom` or `constant` declaration anywhere. The scope is narrow and is stated as such: what is machine-checked is the summation inequality that bounds the ladder, not the estimation of the modules, the rolling-origin construction, or the power simulation — all of which are script computations.

## 5. Conclusions

On locked NCAM M-shift SSB for 1983–2015, last-value persistence is more accurate, on the pooled rolling score, than surplus production, catch-driven stock-flow, autoregressive residuals, lagged initialisation, and prey-informed productivity. Neither prescribed catch treatment reproduced the reconstructed decline in this accounting, though which process accounts for the shortfall is not identified here. Extra structure did not recover the gap to persistence at both horizons, and the lagged-initialisation module enlarged it; the autoregressive module improved on its own declared comparator without closing the gap to persistence.

The strength of that finding is not uniform across modules, and a pre-registered simulation is what separates the cases. Under known ground truth the rule recovers a true autonomous module in 97% of replicates at collapse-window parameters, so the empirical non-retention of that module is evidence about the series rather than about the instrument; for the stock-flow and depensation alternatives at recovery-window parameters power falls below 15%, so their non-retention is weak evidence against those structures. The rule is specific — it declines to retain structure under a persistence-true process in 97–99% of replicates — but an information criterion penalising free parameters would have been the stronger instrument on both axes. Whether a diagnostic can tell an analyst in advance where the rule has power remains unresolved. Scored on a groundwater system in Abaee (2026, Does a one-pool water-balance model improve forecasts of Edwards Aquifer head) and restated under the rule used here, the same criterion withholds retention from a module that beats persistence on the point ranking, so the tie band and the two-horizon requirement are load-bearing rather than nominal. The retained set is empty on both systems, while the series are not pooled and no verdict is transferred between them. The same retention outcome holds on the unpooled xteNCAM series, whose post-2015 stall period fails in the same way the reconstructions stall.

The paper reports a forecast comparison. It does not conclude that the stock is unsustainable, and it does not conclude that the evaluation framework is empirically confirmed on this stock.

The outcome years, the threshold experiment, and the 2022–2023 suspension sharpen the scope statement. The forecast comparison stands as scored; the stock’s decision-relevant uncertainty over 2016–2024 accumulated in the observation and reference-point layer — which vintage reads the state, and which threshold reads the verdict — and in a two-cycle suspension of the observing system itself.

# Part II — Edwards Aquifer, index well J-17

**Source study:** Does a one-pool water-balance model improve forecasts of Edwards Aquifer head? A scored test at J-17. September 6, 2026.

**Prepared in the format of Groundwater (Wiley/NGWA)**

**Article type: Research Paper**

## Abstract

**Problem.** Index-well head forecasting is a recurring operational need, while deliberately simple process-based water-balance models are rarely tested against naive benchmarks under a locked retention rule.

**Approach.** On the 1934–2023 annual-mean J-17 Edwards Aquifer head series, a fixed ladder was scored by rolling and fixed-window out-of-sample RMSE: persistence, training-mean climatology, univariate AR(1), a one-pool stock-flow water balance, residual/delay variants, and climate-informed recharge. A causal module was retained only if it beat both persistence and the next-simpler causal model under a protocol frozen before any score.

**Results.** The stock-flow map that persists last year’s recharge loses to persistence at one year (14.70 vs. 13.23 ft), because annual recharge is near-white ( $\text{corr}(R_t, R_{t-1}) = 0.17$  vs.  $\text{corr}(\Delta H_t, R_t) = 0.74$ ). The univariate AR(1) improves one-year RMSE by 0.39 ft (12.84 ft), but its margin is a coin-flip (mean absolute error a tie, five-year loss, bootstrap interval covering zero). The climatological-flux affine map (M2m) is the best one-step forecaster (12.28 ft) yet is declined by a protocol clause. Given realized fluxes the same map nowcasts (7.55 ft); climate modules gain at most 0.13 ft and none is retained. At five years climatology beats persistence robustly (16.80 vs. 21.11 ft).

**Implications.** For annual J-17 head forecasts persistence and an AR(1) are hard to beat; the one-pool balance, given the year’s fluxes, nowcasts rather than forecasts; multi-year planning should prefer climatology to persisted recharge.

**Keywords:** Edwards Aquifer; J-17 index well; groundwater level forecasting; water balance; forecast evaluation; persistence benchmark; prediction skill

**Article Impact Statement.** Deliberately simple process-based water-balance models are rarely tested against naive benchmarks under a locked retention rule. Across groundwater-forecast evaluation, adding structure should be justified against the forecast that nothing changes. On the 1934–2023 J-17 record, the one-pool balance that persists last year’s recharge loses to persistence at the annual origin (recharge is near-white,  $r = 0.17$ ), while with realized fluxes the same map nowcasts (7.6 ft) rather than forecasts. The transferable lesson is a benchmark-discipline one: climate-independent baseline models are hard to beat at the annual origin in rapidly recharged, institutionally bounded aquifers, and at multi-year horizons the training mean (climatology) is the robust default.

## 1. Introduction

Forecasting aquifer heads at index wells is a recurring operational problem of groundwater management. Drought-stage declarations, springflow protection, and permit adjustments all key on forecasted water levels. The methodological literature has responded with an increasingly rich inventory of groundwater-level forecasting models, from conceptual water balances to data-driven regressions and, prominently, artificial-neural-network and wavelet–neural-network conjunction models trained on measured heads (Daliakopoulos, Coulibaly, and Tsanis 2005; Adamowski and Chan 2011). The field has also begun to institutionalize benchmark culture. GEMS-GER, the first machine-learning benchmark dataset for long-term groundwater levels, standardizes 32 years of weekly observations from 3,207 German wells together with three benchmark models of increasing complexity (Ohmer et al. 2026). Systematic comparisons of nine machine-learning and deep-learning architectures on a karst catchment now anchor the karst groundwater forecasting literature (Zhu et al. 2026).

Three hydrological objects are worth separating, because they are read off the same number. The head record is observed at an access point, the index well. That head indexes a store of water in the aquifer, which is the resource. The store is replenished and drawn by fluxes, which are the flow. The one-pool water balance (a lumped stock-flow model with head change equal to weighted fluxes plus a linear drain) closes this loop approximately. The term “water balance” is used throughout in its increment-structure sense: head change equals weighted fluxes plus a linear drain, with the spring-discharge series stored but deliberately excluded from the

forecasting equation, and a [610, 710] clip standing in for a physical storage floor. The map is not a closed mass balance; this qualification is stated here once and travels with the term.

Whether added structure improves out-of-sample forecasts — as opposed to in-sample fit — is a separate, testable question. The general forecasting literature has made the benchmark discipline explicit: across the M4 competition’s 100,000 series, sophisticated methods did not uniformly beat simple statistical baselines (Makridakis, Spiliotis, and Assimakopoulos 2020). The minimal bar for any proposed module is therefore the forecast that nothing changes — last-value persistence — together with the training mean. Yet the groundwater benchmark studies above compare model families against one another. They do not subject each module to a retention gate against the naive baselines, and none subjects a deliberately simple process-based water balance to a scored ablation against those baselines. This paper supplies that missing test. It applies a scored model-ablation design in which a scored ladder (a forward-ordered set of models evaluated by a fixed retention rule) of incrementally structured one-pool models is evaluated against the two naive baselines. Complexity is kept only if it improves the stated score, decided on out-of-sample error on the predictand itself.

The San Antonio Pool of the Edwards (Balcones Fault Zone) Aquifer, Texas, indexed by the J-17 well, is the natural test bed. Its 1934–2023 daily head record is among the longest managed groundwater series in North America. Its institutional thresholds (the 660-ft Stage I line of the Edwards Aquifer Authority) and its physical threshold (the  $\approx$ 618-ft level at which Comal Springs approaches cessation) are explicit and dated. Recharge and pumpage series exist that are constructed independently of the head series. The aquifer is also a karst system in which regional flow has long been represented — and debated — through equivalent porous media and lumped approaches (Scanlon et al. 2003, for the Barton Springs segment; the lumped-versus-EPM question is inherited by the San Antonio Pool, not settled by that citation). For this reason, the fate of a deliberately simple one-pool water-balance map is a live hydrogeologic question rather than a straw man.

The question is whether stock-flow, residual, delay, or climate-informed recharge modules reduce forecast error of J-17 annual-mean head relative to last-value persistence and to a univariate AR(1). The climate modules enter because El Niño/Southern Oscillation (ENSO) phases shift North American precipitation patterns (Ropelewski and Halpert 1986): September–November Niño 3.4 and lagged climate-division precipitation are precisely the signals observable at an annual forecast origin.

A companion study under separate review applies the same scored design to a marine fishery stock (Northern cod, NAFO 2J3KL); the analogy between the two systems is that the identified driver is not persistent at the forecast origin (Author et al., in review). The two systems’ scores are never pooled, and no retention verdict is transferred between them. This paper does not assess whether the aquifer is sustainable and does not treat springflow or reconstructed storage as co-primary predictands; a two-pool exchange module was specified and not fitted.

## Related work

**Index-well head forecasting.** Forecasting groundwater levels, including at index wells, is a mature modelling area with a substantial literature using statistical, neural-network and process-based approaches on annual and sub-annual series (Daliakopoulos, Coulibaly and Tsanis, 2005; Adamowski and Chan, 2011). That literature is the natural home of this paper.

The departure is the direction of the test. The typical contribution in that area introduces a model and reports its skill, frequently against a persistence or climatological baseline but rarely under a retention rule fixed in advance and then applied without exception. Here the ladder, the baselines and the rule were all specified before any score was read, the rule requires beating *both* persistence and the next-simpler causal model, and a protocol clause can decline the best one-step forecaster — which is what happens to M2m below. The paper’s contribution

is therefore not a new forecasting model; it is a **locked evaluation design applied to a familiar model family**, with the negative result as the finding.

**Simple benchmarks, again.** That simple methods are hard to beat is the recurring result of the forecasting competitions (Makridakis and Hibon, 2000; Makridakis, Spiliotis and Assimakopoulos, 2018). The present paper is a single-system instance of it: persistence and an AR(1) are hard to beat at the annual origin on this series. As in the companion Northern cod study, the scope is one system rather than tens of thousands of series, so the result is a contribution to that phenomenon rather than evidence about method classes in general.

**The mechanism, which is specific to this system.** The reason the one-pool balance fails is not a general deficiency of process-based models but a measurable property of the drivers: annual recharge is close to white ( $r(R_t, R_{t-1}) = 0.17$ ) while the head increment is strongly contemporaneously coupled to recharge ( $r(\Delta H_t, R_t) = 0.74$ ). A model that persists last year’s recharge therefore persists a quantity with almost no memory, which is why it loses to persisting the head itself. This is the same structure the companion study finds on Northern cod, where a weakly autocorrelated surplus production cannot gain either; the mechanisms differ by system, the test does not. Given the year’s realised fluxes the same map nowcasts well (7.55 ft), which is the diagnostic that separates a forecasting failure from a structural one.

**Comparison testing.** The margin comparisons use the standard treatment of predictive-accuracy testing (Diebold and Mariano, 1995); where a margin is reported as a coin-flip, that is the result of the bootstrap interval covering zero rather than an impression.

## 2. Data and Specification

The specification separates data fields from empirical constructs, models, and normative thresholds, so that the scored object is unambiguous.

**Table 1.** The San Antonio Pool specification: J-17 annual-mean elevation, 1934–2023.

| Field                   | Contents                                                                                             | Type  |
|-------------------------|------------------------------------------------------------------------------------------------------|-------|
| System                  | Edwards Aquifer, San Antonio Pool, indexed by J-17                                                   | D     |
| $H_t$                   | Calendar-year mean of daily-high J-17 elevation (ft AMSL)                                            | D     |
| Well                    | TWDB 6837203 / EAA AY-68-37-203                                                                      | D     |
| Domain                  | San Antonio Pool; calendar years 1934–2023                                                           | D     |
| Physical threshold      | Head high enough that Comal Springs does not cease ( $\approx 618$ ft; 1956 daily minimum 612.51 ft) | N / E |
| Institutional threshold | EAA Stage I, 10-day mean J-17 < 660 ft (in force after 2007)                                         | N     |
| Disturbance             | Recharge pulses; unmodeled karst; Uvalde–San Antonio exchange                                        | M     |
| Implementable use       | Pre-EAA pumping; post-1996/2007 permits and critical-period management                               | E     |
| Service series          | USGS 08168710 Comal Springs annual mean discharge (cfs)                                              | D     |

Field types: D = data, E = empirical construct, M = model, N = normative threshold; TWDB = Texas Water Development Board; NSE = Nash–Sutcliffe efficiency. The predictand is a measured well series. It is not a GRACE or G3P storage reconstruction, a MODFLOW or GWSIM inversion, EAA reconstructed storage, the J-27 Uvalde index, San Marcos Springs, or total spring discharge. The predictand is the calendar-year mean of daily *highs*; a daily-mean convention would sit slightly lower and matters near the 660/618-ft lines, and the Authority’s Stage I rule is a 10-day mean — the annual-mean proxy is flagged where used.

Stage I at 660 ft is a 2007 institutional rule and is not applied as if it existed in 1956. Cessation of Comal Springs near 618 ft is a physical service bound. The indicator  $1\{H < 660\}$  on the annual mean is a coarse proxy for the Authority’s 10-day declaration, not the declaration itself.

Annual means use all available daily highs. Years with fewer than 240 observations are dropped. No year falls below the floor (minimum  $n = 242$ , 1939), so the rule never binds on this panel; 1935 ( $n = 258$ ) and 1939 ( $n = 242$ ) are retained as incomplete-coverage means; missing days are not interpolated. The published pre-continuous composite (Beverly Lodges extension) is used as issued. The complete estimation panel ends in 2023, whose values carry provisional TWDB status.

Recharge R is USGS total San Antonio-area recharge ( $10^3$  acre-ft  $\text{yr}^{-1}$ ), estimated by the Puente (1978) stream-loss method and constructed independently of J-17 (Umphres and Choi 2025). Pumpage P is well discharge from Edwards Aquifer Authority Table 1 (Edwards Aquifer Authority 2024/25, covering 1934 onward). R and P are constructed fluxes, not observations of J-17. Total spring discharge is stored and is not used as a driver. Climate predictors, used only in Section 5.4, are September–November Niño 3.4 (HadISST, anomaly relative to 1991–2020) and calendar-year precipitation in Texas climate divisions 06 and 07 (NCEI nClimDiv).

### 3. Forecast Models

The forecast models form a fixed ladder. Each rung adds structure to the previous rung, and each causal rung is scored against the next-simpler rung under the retention rule of Section 4.

**Definition 3.1 (One-pool water-balance map).** With head clipped to  $[610, 710]$  ft, the one-pool map is

$$H_{t+1} = [H_t + \alpha + \beta\tilde{R}_{t+1} + \gamma\tilde{P}_{t+1} + \delta H_t]_{\text{clip}},$$

where  $\tilde{R}$  and  $\tilde{P}$  are the flux values a forecast substitutes for the (unknown) year  $t + 1$  fluxes. Forecasts are issued at the end of year  $t$  from (H, R, P) through  $t$ .

**Table 2.** Model ladder.

| ID        | Class                     | Fluxes at $t + h$           | Role                                                                                                 |
|-----------|---------------------------|-----------------------------|------------------------------------------------------------------------------------------------------|
| persist   | baseline                  | —                           | $\hat{H}_{t+h} = H_t$                                                                                |
| mean      | baseline                  | —                           | training-window mean                                                                                 |
| M1        | autonomous                | —                           | $H_{t+1} = a + \varphi H_t$                                                                          |
| M2        | causal stock-flow         | last $(R_t, P_t)$ persisted | candidate for retention                                                                              |
| M2m       | climatological stock-flow | training-mean $(R, P)$      | affine AR(1)                                                                                         |
| M3        | residual                  | as M2                       | AR(1) residual on $\Delta H$                                                                         |
| M4        | delay                     | as M2                       | symmetry control with the companion fisheries evaluation; not a monitoring constraint of this system |
| M2_oracle | diagnostic                | realized future $R, P$      | excluded from retention                                                                              |

**Remark 3.1 (Class reduction of M2m).** Under constant fluxes  $\tilde{R}_{t+1} = \bar{R}$ ,  $\tilde{P}_{t+1} = \bar{P}$ , M2m reduces to  $H_{t+1} = (1 + \delta)H_t + \text{const}$  and therefore shares M1’s forecast function class — but not its estimator. M2m pins its intercept from the in-sample mean fluxes, an additional identifying use of the recharge and pumpage records in training; the persistence coefficient  $(1 + \delta)$  is not mean-flux dependent. The decline of M2m is a protocol choice recorded in the frozen documents, not a theorem, and it is not part of the retention rule itself (Section 4.1).

Climate-informed variants (Section 5.4) replace persisted R by a one-step forecast of R from information known at  $t$ : AR(1) on R, lagged Niño 3.4, lagged climate-division precipitation, or all three. A precipitation-oracle variant uses year  $t + h$  precipitation and is excluded from retention. For horizons  $h > 1$ , the one-step recharge forecast is held constant.

## 4. Evaluation Design

The evaluation design fixes the scoring rule before any score is computed. The retention rule (a module is kept only if it beats both persistence and the next-simpler causal model) is the gate against which every causal rung of the ladder is judged.

**Definition 4.1 (Primary and secondary scores).** The primary score is RMSE of annual-mean J-17, in feet. Secondary scores are mean absolute error and the Brier score for  $\mathbf{1}\{\hat{H} < 660\}$ , interpreted only for origins at or after 2007; for deterministic 0/1 forecasts this score is a misclassification rate. A sign-hit rate of  $\Delta H$  was listed in an earlier design iteration and is struck: no value is reported for it, and it is not part of any retention decision.

**Definition 4.2 (Retention rule, frozen verbatim).** Let  $\text{RMSE}(\cdot)$  denote the primary rolling-origin RMSE at a fixed horizon  $h$ . A causal module  $M$  is retained only if:

- (H1)  $\text{RMSE}(M) < \text{RMSE}(\text{persist})$ , i.e. it beats last-value persistence, and
- (H2)  $\text{RMSE}(M) < \text{RMSE}(M_{\text{next-simpler causal}})$ , i.e. it beats the next-simpler causal model.

Retention is decided at  $h = 1$  on the point-RMSE comparison; no tie tolerance and no minimum margin is imposed. Diagnostic oracles are excluded from retention. The Comal series is excluded from retention. Retention is decided on the head series only; the service series is scored after that decision is frozen.

### 4.1 Protocol record and deviations

The scoring protocols for the primary pass and the climate pass were frozen and dated (2026-08-25) before the corresponding RMSE tables were computed, and are archived with the analysis code in the public repository. The design is a fixed computational protocol rather than a prospective clinical-style registration; the phrase “pre-registered” is avoided for that reason. The frozen retention rule is Definition 4.2, verbatim above. Three protocol elements sit outside that rule and are therefore recorded here as deviations, in one place:

1. **The M2m class clause.** M2m is listed and then declined on class grounds (Remark 3.1). The clause is in the frozen protocol document, but it is not in Definition 4.2; under the rule as written, M2m satisfies (H1) and (H2) at  $h = 1$ .
2. **The climate-rung comparator.** The frozen Pass-2 protocol document states the climate question as whether a causal recharge forecast reduces primary RMSE on J-17 “relative to persistence and relative to M1” — the retained M1. The comparator is read accordingly: persistence and M1. Declaring the declined M2m as the rung’s (H2) comparator instead would be inconsistent with the frozen document and circular — the gate would be a model the protocol had itself declined. Section 5.4 reports the climate verdict on the declared gate, and the M2m margins are retained as a nested-baseline reading rather than as the gate. The choice changes no frozen verdict — no climate module is retained under either statement — but it fixes the stated mechanism.
3. **The struck sign-hit score** (Definition 4.1).

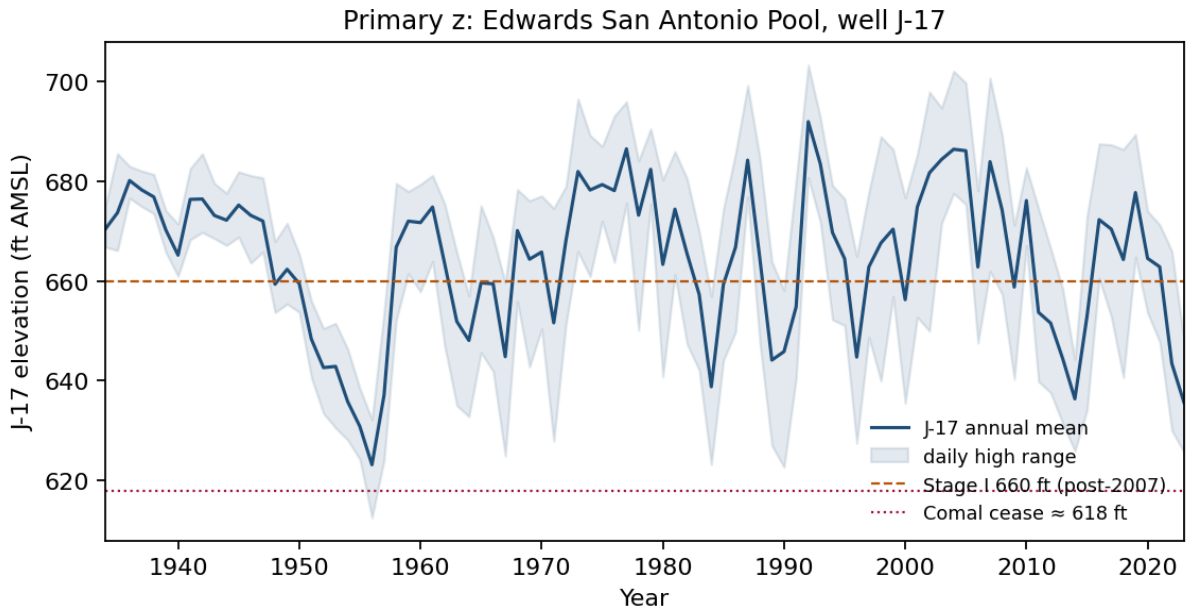
Post-freeze objects, labelled as such: the climate-pass fixed-window scores, the pumpage counterfactuals of Section 5.6, the Comal service-series scoring, and the uncertainty layer of Section 5.3.1 together with its independent replication. None replaces or alters a frozen verdict.

Fixed windows: (1) drought-of-record drawdown, train 1934–1950, test 1951–1956; (2) drought-of-record recovery, train 1934–1956, test 1957–1961; (3) pre-permit wet interval, train 1980–1990, test 1991–1995; (4) critical-period era, train 1997–2014, test 2015–2023.

Rolling origin: minimum 15 training years; horizons  $h = 1$  and  $h = 5$ ;  $n = 75$  and  $n = 71$  origins respectively (every model in Tables 4 and 7 is scored on the identical origin sets; M2m uses the same  $n = 75$  /  $n = 71$ ). The fixed windows train on 16, 22, 10, and 17 transitions respectively and fit four parameters; early rolling origins with the 15-year floor give 14 transitions, so the causal family’s rolling scores carry unstable early-window coefficients (Section 5.2).

## 5. Results

### 5.1 The series



**Figure 7:** J-17 annual mean and daily-high range. 1956 mean 623.15 ft, daily minimum 612.51 ft. 1992 mean 691.96 ft, daily maximum 703.31 ft. 2023 mean 635.68 ft. The annual mean is below 660 ft in 31 of 90 years; the daily minimum is below 618 ft in one year (1956).

### 5.2 Fixed windows

**Table 3.** Fixed-window RMSE (ft). Bold marks the lowest RMSE of the window including the diagnostic oracle.

| Window                  | persist | mean  | M1    | M2           | M2m   | M3    | M4    | oracle       |
|-------------------------|---------|-------|-------|--------------|-------|-------|-------|--------------|
| Drawdown<br>1951–<br>56 | 23.75   | 35.19 | 30.94 | <b>18.11</b> | 27.44 | 18.12 | 18.23 | 19.69        |
| Recovery<br>1957–<br>61 | 43.62   | 14.07 | 56.24 | 55.32        | 37.74 | 55.28 | 55.12 | <b>12.26</b> |

| Window                                | persist | mean  | M1    | M2    | M2m   | M3    | M4    | oracle      |
|---------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------------|
| Pre-permit<br>wet<br>1991–<br>95      | 30.13   | 18.24 | 20.02 | 16.67 | 23.47 | 16.41 | 15.26 | <b>7.18</b> |
| Critical-period<br>era<br>2015–<br>23 | 27.41   | 14.77 | 15.62 | 23.37 | 14.79 | 22.84 | 22.17 | <b>8.70</b> |

Fixed windows are diagnostic, not retention. On the drawdown window the bolded M2 is a continuing-drought artefact, not a forecasting merit: the last observed R is already low, and the map trained on 1934–1950 has the wrong sign on pumpage ( $\gamma = +0.021$ ) — pumping rose as the drought deepened, so least squares aliases the human response into the state transition (simultaneity bias; the short window compounds the identification failure). That identification failure is why M2 “beats” the oracle there. The train-mean baseline is the best non-oracle forecast on the recovery window (14.07 ft) and the critical-period era (14.77 ft), consistent with its rolling five-year win; the residual-persistence rungs M3/M4 track M2 on the drawdown (18.12/18.23 versus 18.11 ft); M4 is the best causal model on the pre-permit wet window (15.26 ft); and both fail with the causal family on the recovery and critical-period windows. The recovery window is also where the [610, 710] clip binds on forecast paths: the M2 trajectory with persisted (R, P) = (43.7, 321.1) runs  $617.1 \rightarrow 611.6 \rightarrow 606.8 \rightarrow 605.3 \rightarrow 605.3$  ft, reaching the 610-ft floor from 1959 onward. The clip never binds on the observed record (annual means 623–692 ft); it binds only on these persisted-drought forecast paths, and it is doing work exactly there.

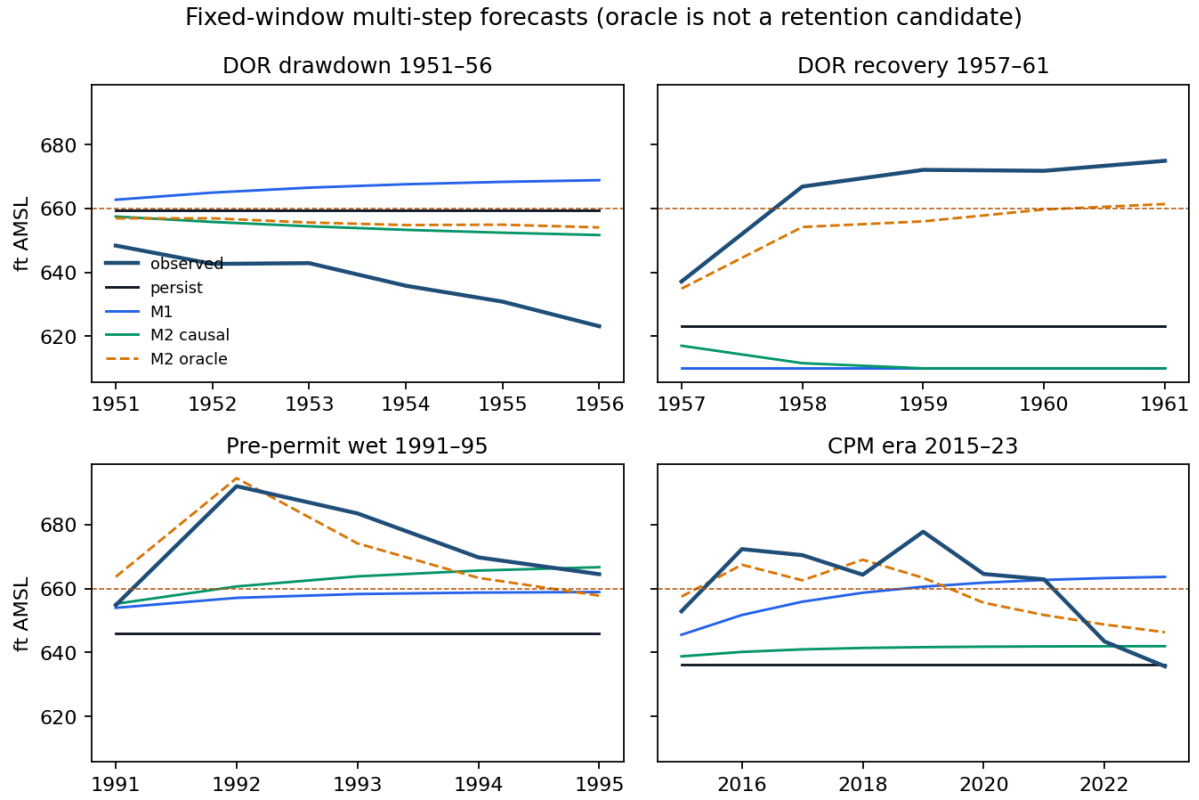
The 1957–61 recovery is a recharge pulse ( $R_{1956} = 43.7$ ,  $R_{1957} = 1142.6 \times 10^3$  acre-ft). Persistence stays at the 1956 floor (RMSE 44 ft). Causal M2 persists drought recharge and falls further (RMSE 55 ft, into the clip). The oracle, given the 1957 flood year, tracks the rise (RMSE 12 ft). Recoveries on this specification are recharge events, not autonomous mean reversion — although the training mean (mean reversion) scores 14.07 ft, second only to the oracle, in this institutionally bounded, rapidly recharged system. The 1992 peak and the 2015–23 window repeat the same split: the oracle follows the recharge year; persistence and persisted recharge do not.

### 5.3 Rolling origin

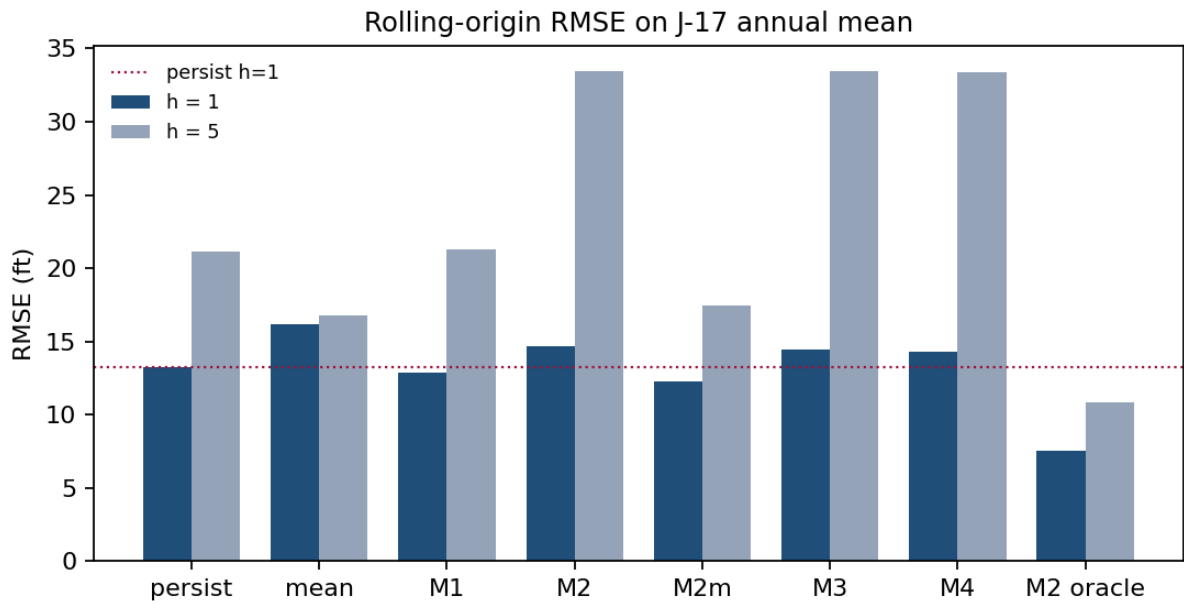
**Table 4.** Rolling-origin summary (ft).

| Model     | h=1 RMSE | h=1 MAE | h=5 RMSE     |
|-----------|----------|---------|--------------|
| persist   | 13.23    | 10.73   | 21.11        |
| mean      | 16.17    | 13.17   | <b>16.80</b> |
| M1        | 12.84    | 10.72   | 21.25        |
| M2        | 14.70    | 11.45   | 33.49        |
| M2m       | 12.28    | 10.22   | 17.44        |
| M3        | 14.46    | 11.12   | 33.46        |
| M4        | 14.30    | 11.17   | 33.39        |
| M2_oracle | 7.55     | 5.79    | 10.86        |

**Table 5.** Retention on rolling  $h = 1$  RMSE (point rule; margins vs persist and vs M1 both shown).



**Figure 8:** Multi-step forecasts from the end of each training window. The oracle (dashed) is diagnostic and uses realized future R, P.



**Figure 9:** Rolling-origin RMSE. The oracle is excluded from retention.

| Model     | vs persist                  | vs M1   | Distinct structure               | Decision                                   |
|-----------|-----------------------------|---------|----------------------------------|--------------------------------------------|
| M1        | $12.84 < 13.23$ ( $-0.39$ ) | —       | output only                      | retained (point rule; margin within noise) |
| M2        | $14.70 > 13.23$ ( $+1.47$ ) | $+1.86$ | causal fluxes                    | reject                                     |
| M2m       | $12.28 < 13.23$ ( $-0.95$ ) | $-0.56$ | no (affine AR(1) function class) | listed; declined by protocol class clause  |
| M3, M4    | worse than persist          | worse   | yes                              | reject                                     |
| M2_oracle | 7.55                        | —       | uses future R, P                 | excluded                                   |

**Retention verdict (rolling  $h = 1$ ).** Applying Definition 4.2 to the  $h = 1$  column of Table 4: M1 satisfies (H1) ( $12.84 < 13.23$ ); as the simplest autonomous model it has no next-simpler causal comparator, so (H2) is vacuous and M1 is retained by the point rule. M2 (14.70), M3 (14.46), and M4 (14.30) each fail (H1). M2m satisfies (H1) ( $12.28 < 13.23$ ) and, against its class-relative comparator M1, also satisfies the (H2) comparison ( $12.28 < 12.84$ ); it is listed and then declined by the class clause of Section 4.1 — a protocol choice, not an outcome of the retention rule. M2\_oracle is excluded as a diagnostic oracle. No stock-flow, residual, or delay module is retained; on the rule as written, the verdict “no stock-flow module is retained” is an artefact of the M2m class clause, since the water-balance-identified affine map with climatological fluxes is the best one-step forecaster tested. What the ladder actually shows is that persisting last year’s recharge is the failure (Section 5.4: persisted-R RMSE 702 versus climatological  $556 \times 10^3$  acre-ft on the recharge target).

M1 is retained by the point-RMSE rule at the one-year horizon. The margin is 0.39 ft on  $n = 75$  and is not a significance claim; its bootstrap interval covers zero (Section 5.3.1), MAE is a tie (10.72 versus 10.73 ft), and at  $h = 5$  M1 (21.25 ft) does not beat persistence (21.11 ft) while the training mean (16.80 ft) beats both. The retention is therefore explicitly a one-year, RMSE-level statement — provisional, a coin-flip recorded by a point rule — and it records a slightly mean-reverting head series, not a confirmation of stock-flow structure. At the decision scale of annual drought-stage declarations the difference is operationally nil. The M2m decline and its 0.56-ft advantage over M1 are likewise within noise (Section 5.3.1): the estimator-level advantage of water-balance identification is visible in the point scores but not separated from zero at this sample size.

**The post-2007 secondary-score record.** Definition 4.1’s second secondary score is scoped to the post-2007 origins, where the 660-ft Stage I rule is in force. On those origins ( $n = 16$  at  $h = 1$ ) the committed records give 660-ft Brier scores of 0.31 (persistence), 0.25 (M1), and 0.19 (the oracle), with the one-year RMSE ranking unchanged (persistence 13.09, M1 12.16, M2 13.31, and the oracle 8.03 ft); at  $h = 5$  ( $n = 12$ ) the M1–persistence ordering reverses on the subsample (17.16 versus 25.10 ft), reported without changing the one-year retention statement. The Brier values are misclassification rates of a coarse annual-mean proxy for the Authority’s 10-day rule on a small subsample, reported from the archived records and not over-read; the full post-2007 sub-sample record ( $h = 1$  and  $h = 5$ ) is archived with the analysis code (`wave_e_edwards/results/rolling_modern_2007.csv`).

### 5.3.1 Uncertainty on the retention margins (post-freeze layer)

A post-freeze uncertainty layer attaches Diebold–Mariano tests (Diebold and Mariano 1995; Newey–West HAC, lag  $h - 1$ ) and moving-block bootstrap intervals (Künsch 1989; block length 8, 10,000 replications, seeded) to every load-bearing margin, computed from the archived per-origin forecast files. It is labelled post-freeze and changes no frozen verdict.

**Table 6.** Uncertainty on the load-bearing margins (post-freeze layer; Diebold–Mariano

with Newey–West HAC, lag  $h = 1$ ; moving-block bootstrap, block length 8, 10,000 replications, seeded).

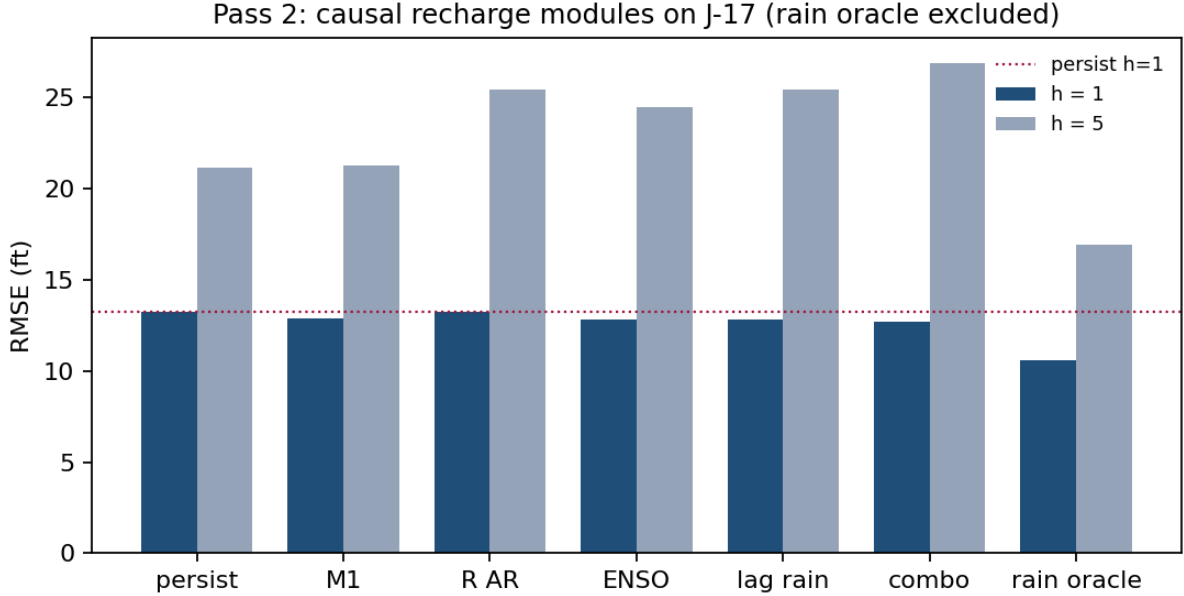
| Comparison<br>(rolling, $h =$<br>1 unless<br>noted) | RMSE margin<br>(ft) | DM statistic | DM p  | 95% block-<br>bootstrap CI<br>(ft) | Separates<br>from zero |
|-----------------------------------------------------|---------------------|--------------|-------|------------------------------------|------------------------|
| M1 – persist                                        | −0.39               | −0.85        | 0.40  | [−1.51,<br>+0.71]                  | no                     |
| M2m –<br>persist                                    | −0.95               | −3.07        | 0.003 | [−1.45,<br>−0.68]                  | <b>yes</b>             |
| M2m – M1                                            | −0.56               | −1.61        | 0.11  | [−1.53,<br>+0.13]                  | no                     |
| M2 – persist                                        | +1.47               | +1.27        | 0.21  | [−0.02,<br>+3.40]                  | no<br>(borderline)     |
| mean –<br>persist ( $h =$<br>5)                     | −4.30               | −1.65        | 0.10  | [−9.72,<br>−2.09]                  | <b>yes</b>             |
| M1 – persist<br>( $h = 5$ )                         | +0.15               | +0.06        | 0.95  | [−5.57,<br>+4.94]                  | no                     |
| M2_combo<br>– M1<br>(climate)                       | −0.13               | −0.21        | 0.83  | [−1.37,<br>+1.31]                  | no                     |
| M2_combo<br>– M2m<br>(nested<br>baseline)           | +0.43               | +0.74        | 0.46  | [−0.56,<br>+1.57]                  | no                     |

Three readings. First, the retention margin itself (M1 – persist, 0.39 ft) is within noise, as is the M2m-over-M1 estimator margin and the M2 causal loss (the last only borderline, its interval touching zero). Under the frozen point-RMSE rule the verdicts stand as frozen; under uncertainty, the only one-year margin that separates from noise is M2m over persistence (−0.95 ft,  $p = 0.003$ ). Second, the five-year climatology win (−4.30 ft) is robust: its bootstrap interval excludes zero. The  $h = 5$  DM statistic ( $p = 0.10$  with HAC lag 4) is known to be undersized under overlapping origins, and the bootstrap interval is the primary evidence. Third, the nested-baseline margin (combo – M2m, +0.43 ft) is itself within noise: the climate rejection is a point-RMSE rule outcome, not a significance finding.

An independent replication of this layer is registered with the repository: the archived script `campaign_e3_dm_uncertainty.py` (in the batch-7 audit directory), a second Diebold–Mariano + moving-block-bootstrap implementation — HAC lag  $h = 1$  with unweighted truncation and population-variance scaling, block length  $h$ , 20,000 replications, seed 0, deterministic (byte-identical on re-execution) — reading the same registered per-origin forecast file. Its verified output (archived alongside it) reproduces every load-bearing reading: the M1 retention margin DM  $z = -0.86$  with block CI [−1.22, +0.56] ft (bootstrap  $p = 0.38$ ); the M2m edge over persistence ( $p = 0.001$ ); the  $h = 5$  climatology win ( $p = 0.035$ ). It also registers two rows this table does not carry — the  $h = 1$  climatology loss (+2.94 ft against persistence,  $p = 0.046$ ) and the M2  $h = 5$  loss (+12.4 ft,  $p < 0.001$ ) — both consistent with the readings above. The two implementations use different seeds, block lengths, HAC weighting, and variance conventions, and agree on every conclusion: the 0.39-ft AR(1) retention margin is statistically indistinguishable from zero, the M2m edge is the only one-year margin that separates from noise, and the five-year climatology win survives.

## 5.4 Climate-informed recharge

Information known at 31 December of year  $t$  comprises  $R_t$ , the September–November Niño 3.4 anomaly of year  $t$ , and calendar-year precipitation in Texas climate divisions 06 and 07. December–February precipitation that includes January of  $t+1$  is not used.



**Figure 10:** Rolling RMSE on J-17 for climate-informed recharge. The precipitation oracle uses year  $t+h$  precipitation and cannot be retained.

**Table 7.** Rolling RMSE, climate-informed recharge (same origin sets as Table 4; margins vs M1 — the frozen Pass-2 gate — and vs the nested M2m baseline both shown for the  $h = 1$  column).

| Model                                    | H, h=1 (ft) | margin vs M1 | H, h=5 (ft)  | R, h=1 ( $10^3$ acre-ft) |
|------------------------------------------|-------------|--------------|--------------|--------------------------|
| persist H /<br>persist R                 | 13.23       | +0.39        | <b>21.11</b> | 702                      |
| M1                                       | 12.84       | —            | 21.25        | —                        |
| M2_Rar                                   | 13.25       | +0.41        | 25.38        | 561                      |
| M2_Renso                                 | 12.82       | −0.02        | 24.42        | 528                      |
| M2_Rprecip                               | 12.80       | −0.04        | 25.38        | 545                      |
| M2_combo                                 | 12.71       | −0.13        | 26.88        | 538                      |
| M2m (the<br>declined nested<br>baseline) | 12.28       | −0.56        | 17.44        | —                        |
| rain<br>climatology                      | —           | —            | —            | 556                      |
| rain oracle                              | 10.56       | —            | 16.91        | <b>354</b>               |

Same-year  $\text{corr}(R, \text{precipitation}) = 0.78$ , which is why the precipitation oracle reduces head RMSE to 10.56 ft; it remains worse than the full  $(R, P)$  oracle (7.55 ft) because the linear rain map misses 1957-scale extremes. That last clause is a result: the oracle gap is not closed by knowing precipitation alone.

Lagged precipitation and September–November Niño 3.4 have modest skill on  $R$  relative to climatology (528–545 versus  $556 \times 10^3$  acre-ft), and they do not constitute forecast structure on head. The frozen Pass-2 gate compares the climate modules against persistence and the

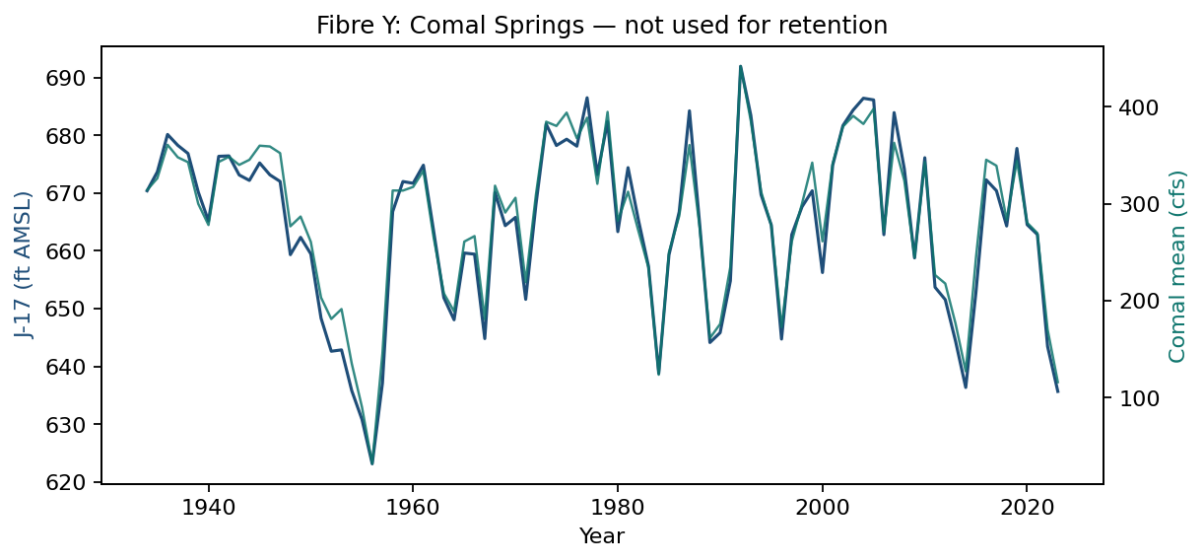
retained M1 (Section 4.1): on that gate the three modules are listed by the point rule — each beats persist and M1 by margins of 0.02, 0.04, and 0.13 ft, all within noise (Section 5.3.1, Table 6) — so the listing is not a skill claim. None is retained, on three stated grounds: the margins against the gate are within noise; at  $h = 5$  all three have RMSE 3–6 ft higher than persistence (the  $h > 1$  climate scores reuse the one-step recharge forecast, held constant over the horizon, so this is a design consequence, not a multi-year climate test); and structurally they are the declined M2m with a weakly adjusted intercept — no additional forecast structure over the retained AR(1) class — so against that nested climatological baseline they lose outright (12.71–12.82 versus 12.28 ft; the combination’s +0.43-ft deficit, Section 5.3.1, also within noise). The rejection is therefore honest with both readings visible: the climate modules beat persistence and the AR(1) by at most 0.13 ft and lose to climatological fluxes; none is retained, and nothing in that verdict is a significance finding. M2\_Rar loses at  $h = 1$  (13.25 ft; 0.41 ft worse than M1) — autoregression on nearly white recharge is not a recharge forecast.

On the 1957–61 recovery, no causal climate-informed module has lower RMSE than persistence (persist 43.6 ft; best causal about 48.8 ft; precipitation oracle 33.7 ft). September–November 1956 is La Niña (−0.92) and does not announce  $R_{1957} = 1143$ .

The remaining fixed windows complete the same record: drawdown 1951–56, M2\_Renso 24.30, M2\_Rprecip 28.67, M2\_combo 24.70, precipitation oracle 18.16 ft; pre-permit wet 1991–95, M2\_Rprecip 22.03, M2\_Renso 23.03, M2\_Rar 23.94, M2\_combo 25.71, oracle 10.98 ft; critical-period era 2015–23, M2\_Rprecip 14.52, M2\_Rar 14.67, M2\_Renso 16.01, M2\_combo 16.57, oracle 9.75 ft — on this one window the two climate-informed modules M2\_Rprecip and M2\_Rar edge past M1 by about one foot (14.52 and 14.67 versus 15.62 ft). On the recharge target itself the fixed-window scores are an order of magnitude coarser on every window (climate modules 199–937; precipitation oracle 80.7–487, against the  $556 \times 10^3$  acre-ft climatology scale of the rolling record), so the marginal head advantage is a window-specific result, not a recharge forecast, and it does not enter the abstract.

Closing the gap between persistence and the oracle would require next year’s recharge, which is not available at the annual forecast origin.

## 5.5 The service series after the retention freeze



**Figure 11:** Comal annual mean versus J-17. Contemporaneous  $r = 0.986$ . The service series is a measured rating curve of the same state — a measured service, not an independent information source.

**The Comal channel (rating-curve reading).** The map  $Q = c_0 + c_1H$  fitted on 1934–

1950 ( $c_0 = -2876$ ,  $c_1 = 4.77$ ) is a linear channel of the same state indexed by J-17. The full-sample contemporaneous correlation is  $r = 0.986$ ; the 1934–1950 fit is quoted because the channel is used on that train, and the full-sample correlation is the redundancy statement.

The channel fails in the drought tail, and the direction matters: at the 1956 annual mean (623.15 ft) it reads  $\approx 97$  cfs against an observed  $\approx 32$  cfs annual mean — it predicts non-cessation — and its zero-discharge level ( $2876/4.77 \approx 602.9$  ft) lies below every observed head (daily minimum 612.5 ft), so the map never predicts cessation where cessation occurred. Near-linear at ordinary heads, the channel fails at cessation exactly where management cares; that tail failure is the one place Comal is not redundant with J-17. One-year Comal RMSE (cfs), constructed by scoring the same ladder directly on the Comal series over the same rolling origins ( $n = 75$ ): persist 71.9, M1 69.0, M2m 68.7, M2 74.8, M3 73.8, M4 73.4, train-mean 89.7, oracle 45.3 — the ranking mirrors the head ranking, as it must for a linear channel.

The service series does not change retention, and it cannot: it is nearly a linear transform of head. Comal is an independent measurement (a USGS spring gauge, not used to construct J-17), but it is not an independent information source. Gravimetric storage or the J-27 Uvalde index would be different objects, not a second observation channel of this specification. Eastern-basin recharge was stored and not used.

## 5.6 Pumpage counterfactuals

A declared scenario layer passes counterfactual pumpage paths through the pre-permit affine map (train 1980–1990, the ladder’s window 3; the short-window identification caveat applies to the map itself), from the observed 1990 head (645.8 ft) over 1991–2023 with the actual recharge sequence. The scenarios are counterfactual simulations of the fitted map — declared scenarios, not forecasts, and no retention implication.

**Table 8.** Pumpage counterfactuals, 1991–2023 (fitted pre-permit map; actual recharge).

| Scenario                   | Mean pumpage ( $10^3$<br>acre-ft yr $^{-1}$ ) | End head 2023 (ft) | Minimum head (ft) | RMSE vs observed<br>heads (ft) |
|----------------------------|-----------------------------------------------|--------------------|-------------------|--------------------------------|
| actual                     | 382.1                                         | 641.6              | 640.2             | 8.56                           |
| pumpage<br>frozen at 1990  | 489.4                                         | 630.9              | 630.5             | 14.22                          |
| pre-permit<br>mean pumpage | 469.8                                         | 632.4              | 632.0             | 13.02                          |
| 20% pumpage<br>cut         | 305.7                                         | 646.8              | 645.5             | <b>7.19</b>                    |

The observed 2023 head is 635.7 ft. Four readings follow. First, pumpage is a secondary lever in this map: the full spread of counterfactual policies — from a 20% cut below actual to freezing at the 1990 peak — spans 630.9–646.8 ft at 2023, and the map’s own RMSE against the observed record (7.2–14.2 ft) is of the same order as the policy spread itself; the 5-ft effects below are within model error and are not policy estimates. Second, the 20%-cut path is the closest to the observed record of the four scenarios (7.19 versus 8.56 ft) — an artefact of the map running high on this window (the actual-pumpage path ends 5.9 ft above the observed 2023 head), not evidence that the Authority’s pumpage was near-optimal; the counterfactuals do not repair the ladder’s primary failure, which Section 5.3 traces to the recharge series’ persistence failure. Third, the direction is right and the magnitude modest: the 20% cut raises the simulated 2023 head by 5.2 ft (646.8 – 641.6) relative to the actual-pumpage path. Fourth, the recharge coefficient dominates the map through its range, not its magnitude: a  $1000 \times 10^3$  acre-ft recharge difference moves the head by roughly 17.5 ft at the pre-permit coefficient ( $\hat{\beta} \approx 0.0175$  on the 1980–1990 train; the pumpage coefficient on that train is  $-0.031$  ft per  $10^3$

acre-ft, larger per unit but acting on a range of  $\sim 300\text{--}540$  against recharge’s 44–2,486), and  $\gamma$  is an ordinary-least-squares association the paper itself diagnoses as simultaneity-contaminated (Section 5.2), not a policy lever. The full-sample coefficients  $(\hat{\beta}, \hat{\gamma}) = (0.017, -0.026)$  differ from the pre-permit window’s because the windows differ.

## 6. Discussion

The one-pool increment  $\Delta H$  tracks recharge ( $r = 0.74$ ), and the oracle’s RMSE is 43% below persistence at  $h = 1$  and 49% below at  $h = 5$ : the map accounts for contemporaneous increments when the year’s water is known. As a one-year forecast, causal stock-flow fails because the dominant increment is not persistent; residual and delayed variants inherit that timing error. Recoveries in 1957 and 1992 are missed by every causal model and captured by the oracle. The identified driver has the right timing and is unforecastable one year ahead — the same pattern a companion evaluation finds on Northern cod, where a more accurate catch series does not rescue constant-productivity surplus production (Author et al., in review). The two score tables are not pooled.

A second question is why the AR(1)’s edge over persistence is as small as it is, and here the fitted head autocorrelation is the operative quantity. An estimated AR(1) with coefficient  $\rho$  nests persistence at  $\rho = 1$ ; as the head series approaches that limit the two forecasts converge, and the AR(1)’s advantage shrinks toward the noise in estimating  $\rho$  itself. On this record the head is persistent but not nearly a unit root (annual  $AC(1) = 0.64$ ), which leaves the AR(1) a real but slim 0.39-ft margin — consistent with the coin-flip diagnostics reported above. The prediction implied is directional and cheap to state: on a more strongly autocorrelated head record the AR(1) should beat persistence by *less*, not more, and at sufficiently high  $AC(1)$  it should lose. An exploratory port of this ladder to a second, separately regulated pool of the same aquifer (annual head  $AC(1) = 0.84$ ) is consistent with that direction — there the AR(1) loses to persistence by 0.13 ft — but a single additional pool is a direction, not a test, and no claim is made here beyond the mechanism. The point for the present result is that the AR(1)’s retention is a weak-autocorrelation phenomenon and should not be read as evidence that an output-only model is generally preferable.

Two features distinguish this design from the groundwater benchmark literature it engages. First, the retention gate: GEMS-GER ships three benchmark models and reports the fraction of wells for which the best one reaches  $NSE > 0.5$  (Ohmer et al. 2026), and the karst benchmark of Zhu et al. (2026) ranks nine architectures by RMSE and  $R^2$  — both compare model families against one another, while here every module must beat persistence and the next-simpler causal model under a rule frozen before scoring, and on that rule no causal module is retained at the one-year horizon (the climate rungs pass the frozen Pass-2 gate by within-noise margins and are declined on class grounds; Section 5.4). Second, the horizon contrast: no benchmark study of this basin reports a training mean beating persistence at a longer horizon. The five-year climatology result (16.80 versus 21.11 ft) is the specification’s most directly transferable candidate for management planning that keys on multi-year outlooks, with the scope stated: the San Antonio Pool is a rapidly recharged, institutionally bounded system, so the training mean is informative at long horizons here, and a non-stationary fossil aquifer under sustained depletion would not sustain a mean-reverting baseline.

Two specification points carry over to the interpretation. First, next-year pumpage is partially an institutional scenario — permits and critical-period rules respond to head — rather than a purely exogenous flux like rainfall; persisting last year’s pumpage is a declared simplification, and the oracle pumpage is in no information set. Second, the karst setting matters. The annual affine one-pool specification is not a karst model: conduits, the Uvalde–San Antonio divide, unconfined recharge-zone storage, and the confined-zone pressure response remain in the residual, and the long-standing question whether lumped or equivalent-porous-media representations can carry regional flow (Scanlon et al. 2003) is inherited, not resolved, by this design.

What the design shows is that at the annual origin, the information carried by such a map is timing-bound: with realized fluxes the same map is a nowcast of the year’s head (7.55 ft RMSE), and the contemporaneous increment  $\Delta H_t$  against  $R_t$  ( $r = 0.74$ ) is the closure statement; the word “certificate” is retired for both. The retained AR(1) admits a complementary reading: to first order, spring discharge obeys Darcy proportionality to head above the spring level, and the autonomous solution of that drainage law is an affine autoregression  $H_{t+1} \approx (1 - k)H_t + kH_s$ ; the fitted  $\hat{\varphi} = 0.66$  (the M1 estimate; the full-sample Pearson correlation  $\text{corr}(H_t, H_{t-1}) = 0.64$  is a different quantity) is consistent with a drainage-decay coefficient  $\hat{k} \approx 0.34 \text{ yr}^{-1}$ . The module is output-only in the protocol’s sense — no flux data enter the forecast — but the shape it fits carries the aquifer’s own free-drainage momentum; this is an interpretation in discussion, not a retention reason.

The wrong leading indicator is worth recording. The 1950s decline is not a Stage I event: the 660-ft line is a 2007 rule, and slack to 660 ft is the wrong leading indicator for 1951–56, just as the 2016 Northern cod limit reference point was the wrong leading indicator for 1983–90 in the companion study. 1956 is a physical near-cessation at Comal (annual mean springflow 32 cfs; daily J-17 minimum 612.51 ft).

A two-pool exchange module was specified and not fitted; no barrier or exchange term was fitted. Post-1997 pumpage no longer spikes with drought as in 1956 ( $321 \times 10^3$  acre-ft from wells); that change is already in  $P_t$ , and a separate critical-period switch adds no degree of freedom beyond the pumpage series.

Limitations follow the data and the map. Total-area  $R$  and  $P$  mix the San Antonio and Uvalde pools; recharge is a Puente estimate; pumpage includes unreported domestic, livestock, and federal use. Neither series is head:  $R$  and  $P$  are constructed fluxes, not observations of J-17. M4 is a one-year information delay and a symmetry control with the companion fisheries evaluation — J-17 is a telemetered gauge whose head is recorded daily, so the module prices a theoretical information lag rather than a constraint of this system’s monitoring. The sample is 90 years with four short test windows, and the 0.39-ft AR(1) margin is not a significance claim (Section 5.3.1: its interval covers zero). Additional climate indices (PDO, AMO) were not run; they would enter through the same one-step recharge forecast and cannot alter the timing result. A mid-year nowcast would require a new evaluation protocol.

## 7. Conclusions

*The general finding, stated before the basin that demonstrates it.* Process-based models are not self-justifying. A one-pool water balance is more mechanistic than last-value persistence, and that added mechanism is widely treated as a reason to prefer it. It is not: it is a *claim*, testable against a naive benchmark, and this paper’s general contribution is a statement of when such claims fail and why. The value of a process model is bounded by the predictability of the driver it persists. Where the model’s causal content consists of carrying a driver forward in time, and that driver is close to white, the structure adds parameters without adding signal — the mechanism is real but carries no forecast information at the horizon in question, and persisting it is overhead rather than insight. That is a statement about evaluation design, not about hydrology: *a model that persists a near-white driver has no claim to beat persistence, whatever its mechanistic fidelity.* It replicates in the companion Northern cod study (Abaee, 2026), where scored surplus-production structure likewise failed to beat last-value persistence at the annual origin. The underlying mechanism differs by system — a near-white annual recharge here, a weakly autocorrelated surplus there — but the test does not, and that invariance is the point: each system is held to the standard that added structure clears a naive benchmark, and the verdict is allowed to be negative. Two consequences follow that are general even where the specific numbers are not. First, a negative result of this kind is informative only against a protocol fixed in advance, which is why the retention rule here was frozen before any score was computed. Second, the failure is *diagnostic* rather than merely negative: it locates the defect in

the driver’s spectrum, telling the modeller where to look rather than only that the model lost.

*The demonstration.* On locked J-17 annual-mean head, last-value persistence is more accurate than a causal one-pool balance that persists last year’s recharge, because annual recharge is near-white ( $r = 0.17$ ) while the contemporaneous increment tracks it ( $r = 0.74$ ). The univariate AR(1) improves one-year RMSE by 0.39 ft and is retained as an output-only model by the frozen point rule; the margin is a coin-flip (MAE tie, five-year loss, bootstrap interval covering zero), and the same affine class fitted through the water balance with climatological fluxes does better still (12.28 ft, the only one-year margin separated from noise) but is declined by a protocol class clause outside the retention rule. The same balance, given realized recharge and pumpage, nowcasts the current year (7.55 ft, a 43% reduction); climate variables known at the annual origin do not recover that gap; and at five years, climatology wins, robustly (16.80 versus 21.11 ft, interval excluding zero).

*Why this basin is the demonstration and not the claim.* The J-17 record is the test bed, and the replication in the companion study is what makes the finding general rather than local. Nothing below should be read as transferring the J-17 numbers themselves to another basin.

*The replication, stated in one place.* The transferable lesson stated at the head of this section — that a deliberately simple process-based model must be justified against naive persistence and the training mean, and may fail to earn that justification when the driver it persists is near-white — replicates in the companion Northern cod forecast study (Abaee 2026), where scored surplus-production structure likewise failed to beat last-value persistence at the annual origin. The test is the same in both systems even though the mechanism is not. That is the general content of the paper; the J-17 basin is its demonstration, and the cod series is its replication.

*Practical reading.* Two choices follow for a manager. At the annual origin, do not prefer a one-pool water-balance forecast over persistence on timing grounds — the map is only as good as the year’s realized fluxes, which are not known at the forecast origin. For multi-year planning, prefer climatological baselines to persisted recharge: the five-year training-mean result (16.80 versus 21.11 ft) beat persistence with an interval excluding zero, and it is the most directly transferable candidate for a district that keys on multi-year outlooks. The pumpage counterfactuals (Section 5.6) add a third, caveated reading: pumpage is a secondary lever on this map (the 20% cut raises the simulated 2023 head by 5.2 ft, but the full policy spread is within the map’s own error band), so pumpage regulation is not a substitute for forecasting skill. A fourth reading concerns the AR(1) itself: its retention here is a weak-autocorrelation phenomenon (Section 6), so a district whose head record is more strongly autocorrelated than J-17’s should expect the AR(1)’s edge over persistence to be smaller, not larger, and should re-score rather than inherit this verdict.

*Generalizability boundary.* The class of systems for which the finding is expected is stated explicitly: a rapidly recharged, institutionally bounded aquifer, not a non-stationary fossil aquifer under sustained depletion. Conducts, the Uvalde–San Antonio divide, unconfined recharge-zone storage, and the confined-zone pressure response remain in the residual for this one-pool map (Section 6), and the single-well test bed is the San Antonio Pool’s well J-17 only; no transfer to the Uvalde Pool (J-27) is claimed, consistent with the programme’s no-pooling discipline. The Uvalde figure quoted in Section 6 is used only to state the direction of a mechanism, not to score a model: no Uvalde result enters the retention verdicts, the two records are not pooled, and that exploratory comparison is not offered as a test of anything claimed here.

The paper reports a forecast comparison. It does not conclude that the aquifer is unsustainable, and it does not conclude that the evaluation framework is empirically confirmed on this basin.

## Cross-system conclusion

The two studies above were designed to answer one question with two systems, and the answer is the same on both.

*Added structure did not earn its keep at the annual origin.* On the Northern cod series no module is retained on either unpooled specification: persistence is more accurate than surplus production, catch-driven stock-flow, autoregressive residuals, lagged initialisation, and prey-informed productivity. On the Edwards J-17 series, persistence is more accurate than a causal one-pool balance that persists last year’s recharge, and the one module that is retained — the univariate AR(1) — is retained by a margin the study itself shows to be a coin-flip. Neither result says the models are wrong as descriptions. Both say that, at the horizon at which they were scored, their structure did not buy accuracy over the assumption that next year looks like this year.

*The failure is diagnostic, not merely negative.* This is the part that generalises, and it is why the paper reports a negative result rather than a shrug. In each system the reason added structure fails is located in the spectrum of the driver the model persists: a weakly autocorrelated surplus production in the fishery, a near-white annual recharge ( $r = 0.17$ ) in the aquifer. In both cases the model is persisting a quantity that does not persist, so its causal content cannot contribute at the forecast origin no matter how faithfully it is specified. That converts “the model lost” into “the model lost *here, for this reason*”, which is actionable: it tells the modeller to look at the driver’s predictability before adding structure that depends on it, and it predicts that the same structure will succeed on a system whose driver is more persistent. The cod study makes the complementary point in the positive direction — under known ground truth the rule recovers a true autonomous module in 97% of replicates — so the instrument is not biased against structure; the structure genuinely did not earn retention.

*The protocol is what makes the negative result informative.* A finding that a model family fails is worth little unless the rule was fixed before the scores were seen, and both studies freeze the scoring core in advance. Both also report power, because a non-retention can mean either “the structure is absent” or “the test could not see it”: on the cod series the distinction is decisive (97% against below 15%, depending on the alternative), and reporting it is what separates the cases. Both studies decline to pool across systems, so neither verdict is diluted by the other.

*What transfers and what does not.* What transfers is the test — that a process-based model must clear naive persistence and the training mean under a protocol fixed in advance, and that the driver’s predictability bounds what it can achieve — together with the diagnostic reading of a failure when it comes. What does not transfer is any specific number: the retention verdicts, the margins and the power figures belong to their systems and are reported separately throughout. A reader whose system has a more strongly autocorrelated driver than either of these should expect a different verdict and should re-score rather than inherit one; both studies say so explicitly.

*The general lesson, in one sentence.* Mechanistic fidelity is a hypothesis about accuracy, not a substitute for measuring it — and when a model’s mechanism amounts to persisting a near-white driver, that hypothesis is testable, and in two unrelated systems it failed.

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## Declarations

### Data availability

All input data, analysis scripts, result files, and the frozen model specification are archived at <https://github.com/MIKEAA2020/general-sustainability> and <https://zenodo.org/records/22553609>. The outcome-year, vintage, and threshold analyses of Sections 3.8–3.10 use the RAM Legacy Stock Assessment Database v4.66 (10.5281/zenodo.14043038; stock COD2J3KL), the digitized NCAM Table A2 and xteNCAM Table 17 files, and the capelin acoustic index; these are archived in-repo under `paperE1_calibration_data_v1_wave_e_cod/` with locked provenance, together with the calibration record `paperE1_calibration_data_v1.md`.

### Reproducibility

All computations are deterministic: within a fixed interpreter and library stack, repeated executions reproduce every result file byte for byte. An independent re-execution verified all 29 recorded output checksums and regenerated all 30 result files byte-identically; one registered file carries no pinned checksum and is covered by the regeneration comparison. Across environments, the intervention runners and the OLS-based prediction runners regenerate their outputs byte for byte, while the four optimizer-based forecast runners (L-BFGS-B fits) reproduce every scored row at printed precision with one exception: the M1b rolling row on Specification B, whose Allee optimum is environment-sensitive at the  $\pm 17$  kt level ( $h = 1$ : 151.6 versus 153.2 kt;  $h = 5$ : 445.5 versus 462.5 kt, on Python 3.12/numpy 2.1.3/scipy 1.14.1 against the recorded original environment). That sensitivity is consistent with the identification fragility reported for M1b in Section 3.1 ( $s$  descending towards zero,  $K$  resting at the 500-kt multi-start initialiser), and it is immaterial to the retention verdict, which persistence wins at both horizons by margins far larger than the instability. The archive records the checksums of the original-environment outputs; the deterministic-in-environment claim and this sensitivity note together are the reproducibility statement. Sections 3.8–3.10 are produced by `paperE1_cod_forecast_ladder_v59_verification.py` (standard library only), which regenerates every number in those sections from the archived data files and asserts the cross-checks recorded there.

The uncertainty layer of Section 3.5 is produced by a seeded, deterministic script applying Diebold–Mariano (HAC) statistics and a Künsch moving-block bootstrap to the archived per-origin forecast files; the persistence baseline is recomputed there on the identical origin sets from the registered series and asserted against the recorded origin-matched values. The script and its output are archived with the repository.

The section scripts are archived at `wave_e_cod/src/` in the repository:

```
python3 wave_e_cod/src/run_ladder.py
python3 wave_e_cod/src/run_xte.py
python3 wave_e_cod/src/run_capelin_regime.py
```

```
python3 wave_e_cod/src/run_capelin_index.py
python3 wave_e_cod/src/compare_catch.py
python3 wave_e_cod/src/make_figures.py
```

## CRediT authorship contribution statement

A.A. conceptualized the entire work, wrote, reviewed and edited the manuscript.

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None.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Code availability

The scored ladder and every auxiliary campaign are produced by the scripts named above, archived with their outputs. The verification script `paperE1_cod_forecast_ladder_v60_verification.py` re-runs the record and recompares the printed scores, reference points and window facts against the locked input files; it is archived alongside this manuscript.

**Edwards Aquifer data availability.** All input data, analysis scripts, and result files are archived in the public repository at <https://github.com/MIKEAA2020/general-sustainability>, together with the frozen scoring protocols (dated 2026-08-25, locked before any score was generated). J-17 daily highs: Texas Water Development Board well 6837203. Recharge: Umphres and Choi (2025), USGS data release, <https://doi.org/10.5066/P1BI62NY>. Pumpage: Edwards Aquifer Authority Table 1. Comal Springs: USGS 08168710. Niño 3.4: NOAA PSL HadISST (raw file registered with the repository). Precipitation: NCEI nClimDiv — the raw file is not distributed with the repository (provenance URL archived in the sources index), so the three precipitation columns of the fixed panel are not reproducible from the registered code alone, while the two Niño columns rebuild from the registered file to machine precision; scoring from the registered analysis panel does not require the nClimDiv file. The registered twenty-column analysis panel is the dataset of record for all scored analyses. All computations are deterministic: re-executing the registered scripts in a fresh environment regenerated every archived result file byte for byte, and all scored rows recompute from the per-observation forecast files and the registered series. The pumpage counterfactual layer (Section 5.6) is produced by `rerun_campaigns/campaign_e3_pumpage_scenarios.py`, archived alongside its outputs, and regenerates them exactly. The post-freeze uncertainty layer (Section 5.3.1) is produced by `wave_e_edwards/src/e3_audit_uncertainty.py` (seeded, deterministic; Diebold–Mariano with Newey–West HAC and moving-block bootstrap on the archived per-origin forecast files), with its outputs archived as `wave_e_edwards/results/e3_audit_uncertainty.json`; the clip-binding statement of Section 5.2 is computed there from the registered panel and reproduces both fixed-window M2 RMSEs (18.11 and 55.32 ft) exactly. The independent replication of the post-freeze uncertainty layer is registered as `batch 7 (audits of agent arena 1 paper rewrites)/campaign_e3_dm_uncertainty.py` (archived), with its deterministically reproduced output archived alongside it at `batch 7 (audits of agent arena 1 paper rewrites)/results/e3_dm_uncertainty.csv`; it recomputes the Diebold–Mariano and moving-block-bootstrap table from the same registered per-origin forecast file under different seeds, block lengths, and HAC conventions, and every load-bearing conclusion of Section 5.3.1 is unchanged under either implementation.

**Edwards Aquifer code availability.** The scored ladder, the post-freeze uncertainty layer and the pumpage counterfactuals are produced by the scripts named above, archived with their outputs. The verification script `paperE3_edwards_forecast_ladder_v17_verification.py` re-runs the ladder from the registered panel and recompares every cell of the corresponding Edwards-source tables (numbered 3, 4 and 7 in the standalone source) against the recomputed scores; it is archived alongside this manuscript.

## **AI declaration**

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.